MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, ADVANCED LOW-POWER SCHOTTKY TTL, FLIP-FLOPS, MONOLITHIC SILICON

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, Advanced low-power Schottky TTL, flip-flops, bistable logic microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Hex D-type flip-flop with clear
02	Quadruple D-type flip-flop with clear
03	Octal D-type flip-flop with transparent latch and 3 state outputs
04	Octal D-type flip-flop with 3 state outputs

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
E	D-2 (16-lead, 1/4" x 7/8"), dual-in-line package
F	F-5 (16-lead, 1/4" x 3/8"), flat package
R	D-8 (20-lead, 1/4" x 1-1/16"), dual-in-line package
S	F-9 (20-lead, 1/4" x 1/2"), flat package
2	C-2 (20 terminal, .350" x .350"), square chip carrier package.

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to +7.0 V dc
Input voltage range	-1.5 V dc at -18 mA to +7.0 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation (P_D) 1/	
Device type 01	104.5 mW
Device type 02	77 mW
Device type 03	148.5 mW
Device type 04	170.5 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC}):	
Case E	50°C/W
Case F	70°C/W
Case R	40°C/W
Cases S and 2	60°C/W
Junction temperature (T_J)	+175°C

1/ Must withstand the added P_D due to short circuit test (e.g., I_{OS}).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions.

Supply voltage	4.5 V dc minimum to 5.5 V dc maximum
Minimum high-level input voltage (V_{IH})	2.0 V dc
Maximum low-level input voltage (V_{IL})	0.8 V dc
Normalized fanout (each output) 2/	10 maximum at low logic level 20 maximum at high logic level
Case operating temperature range (T_C)	-55°C to +125°C
Minimum width of Clock pulse (t_p(clk))	
type 01, 02 25°C	10 ns
-55/125°C	12.5 ns
type 04 25°C	14 ns
-55/125°C	16.5 ns
Minimum width of Clear pulse (t_p(clear))	
type 01, 02	15 ns
Minimum width of Enable pulse (t_p(Enable))	
type 03	10 ns
Minimum setup time before clock (t_{setup})	
type 01, 02	15 ns
type 04	10 ns
Minimum hold time after clock (t_{hold})	
type 01, 02	0 ns
type 04	4 ns
Minimum setup time before Enable (t_{setup})	
type 03	10 ns
Minimum hold time after enable (t_{hold})	
type 03	7 ns
Minimum clear inactive state time before clock	
type 01, 02	8 ns

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

2/ The device shall fanout in both high and low levels to the specified number of inputs of the same device type as that being tested.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions 1/ $-55°C \le T_C \le +125°C$		Device type	Limits Min	Limits Max	Unit
High-level output voltage	V_{OH}	V_{CC} = 4.5 V	I_{OH} = -400 µA	01,02	2.5	---	V
		V_{IN} = 2.0 V, V_{IL} = 0.8 V	I_{OH} = -1.0 mA	03,04	2.4	---	
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V	I_{OL} = 4 mA	01,02	---	0.4	V
		V_{IH} = 2.0 V, V_{IL} = 0.8 V	I_{OL} = 12 mA	03,04	---	0.4	
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{OL} = -18 mA	T_C = 25°C	All	---	-1.5	V
Low-level input current	I_{IL}	V_{CC} = 5.5 V, V_{IN} = 0.4 V		01,02 03	0	-100	µA
				04	0	-200	
High-level input current	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.7 V		All	---	20	µA
	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 7.0 V		All	---	100	µA
Output current, low level, outputs disabled	I_{OZL}	V_{CC} = 5.5 V, V_O = 0.4 V		03,04	---	-20	µA
Output current, high level outputs disabled	I_{OZH}	V_{CC} = 5.5 V, V_O = 2.7 V		03,04	---	20	µA
Short circuit output current	I_O 1/	V_{CC} = 5.5 V, V_{IN} = 2.25 V		All	-30	-112	mA
Supply current	I_{CC}	V_{CC} = 5.5 V, V_{IN} = 4.5 V 2/		01		19	mA
				02		14	
Supply current outputs high	I_{CCH}	V_{CC} = 5.5 V, V_{IN} = 4.5 V		03		16	mA
				04		19	
Supply current outputs low	I_{CCL}	V_{CC} = 5.5 V, V_{IN} = 0 V		03		25	mA
				04		28	
Supply current outputs disabled	I_{CCZ}	V_{CC} = 5.5 V		03		27	mA
				04		31	
Maximum clock frequency	f_{MAX}			01,02		40	MHz
				04		30	

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ $-55°C \le T_C \le +125°C$	Device type	Limits Min	Limits Max	Unit
Propagation delay time, low-to-high, CLK to Q	t_{PLH1}	V_{CC} = 5.0 V C_L = 50 pF R_L = R1 = 500Ω	01,02	3	17	ns
			04	3	15	
Propagation delay time, high-to-low, CLK to Q	t_{PHL1}		01,02	5	20	ns
			04	5	18	
Propagation delay time, low-to-high, D to Q	t_{PLH2}		03	2	14	ns
Propagation delay time, high-to-low, D to Q	t_{PHL2}		03	4	19	ns
Propagation delay time, low-to-high, ENC to Q	t_{PLH3}		03	6	26	ns
Propagation delay time, high-to-low, ENC to Q	t_{PHL3}		03	7	27	ns
Propagation delay time, low-to-high, Clear to $\overline{Q}$	t_{PLH4}		02	5	20	ns
Propagation delay time, high-to-low, clear to Q	t_{PHL4}		01,02	8	26	ns
Propagation delay time, disable to high level	t_{PZH}		03	5	24	ns
			04	5	19	
Propagation delay time, disable to low level	t_{PZL}		03	6	22	ns
			04	7	20	
Propagation delay time, high level to disable	t_{PHZ}		03	2	16	ns
			04	2	12	
Propagation delay time, low level to disable	t_{PLZ}		03	2	12	ns
			04	3	24	

1/ The output conditions have been chosen to produce a current that closely approximates one-half the true short-circuit current, I_{OS}.

2/ I_{CC} is measured with D inputs and $\overline{CLR}$ grounded.

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3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Logic diagrams and terminal connections. The logic diagrams and terminal connections shall be as specified on figure 1 and figure 2.

3.2.2 Truth tables. The truth tables shall be as specified on figure 3.

3.2.3 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained and available upon request.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.4 herein.

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 10 (see MIL-M-38510, appendix E).

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,7,9,10,11	1*,2,3,7,9
Group A test requirements (method 5005)	1,2,3,7,8,9,10,11	1,2,3,7,8,9
Group B test requirements (method 5005) subgroup 5	1,2,3,9,10,11,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3
Additional electrical subgroups for group C periodic inspections	N/A	10,11
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

*PDA applies to subgroup 1 (see 4.2c).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in (method 1015 of MIL-STD-883).

(1) Test condition D, using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Electrical test requirements shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical test requirements shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D, using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

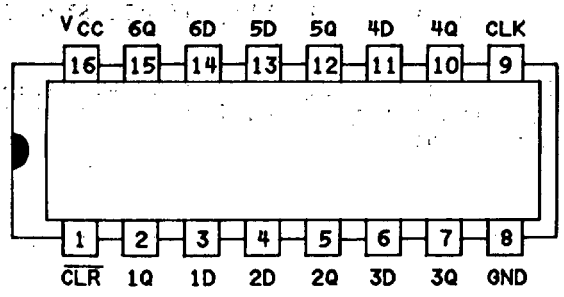
c. Subgroups 3 and 4 shall be added to group C inspection for class B devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

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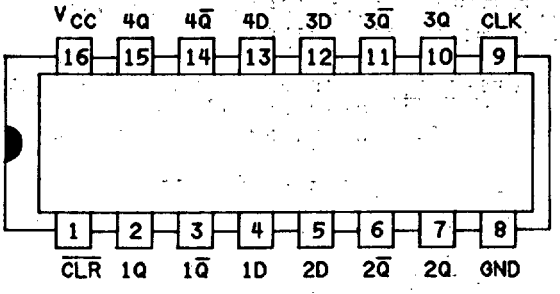
Device type 01

Cases E and F



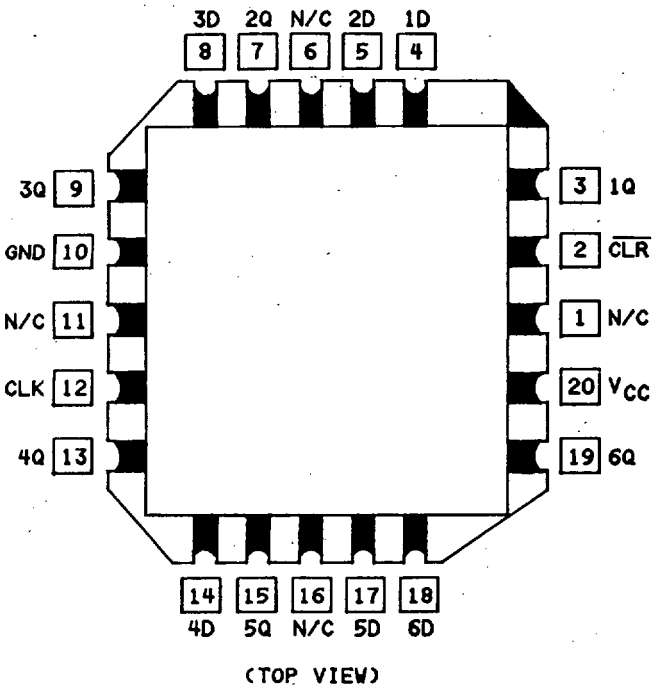
Device type 02

Cases E and F



Device type 01

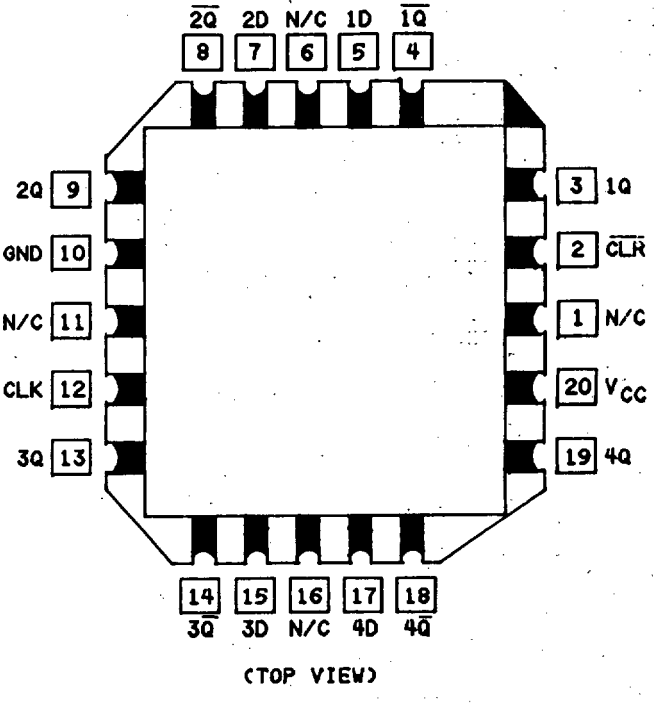
Case 2



(TOP VIEW)

Device type 02

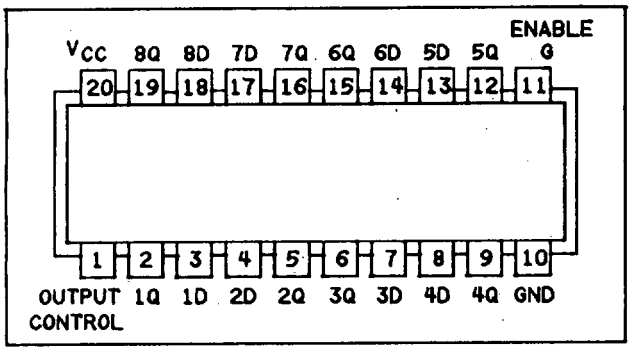
Case 2



(TOP VIEW)

Device type 03

Cases R, S, and 2



Device type 04

Cases R, S, and 2

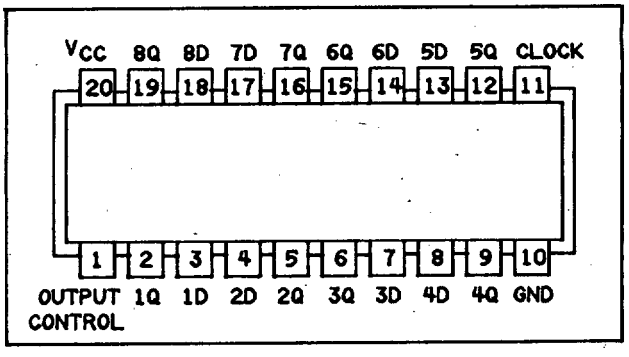
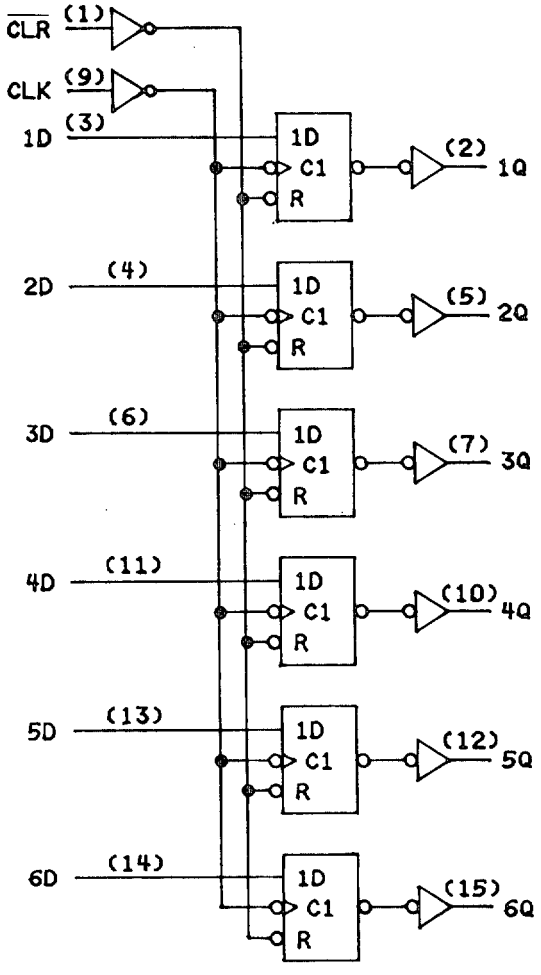


FIGURE 1. Terminal connections.

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Device type 01



Device type 02

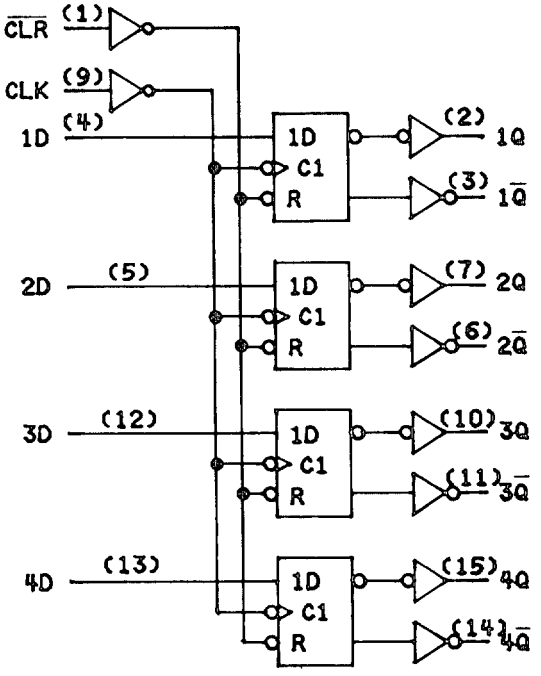
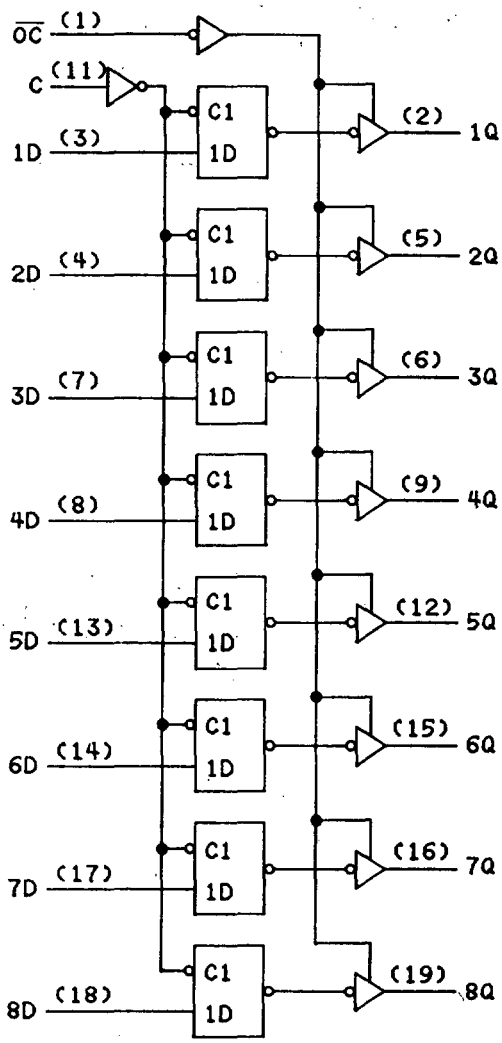


FIGURE 2. Logic diagram.

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Device type 03



Device type 04

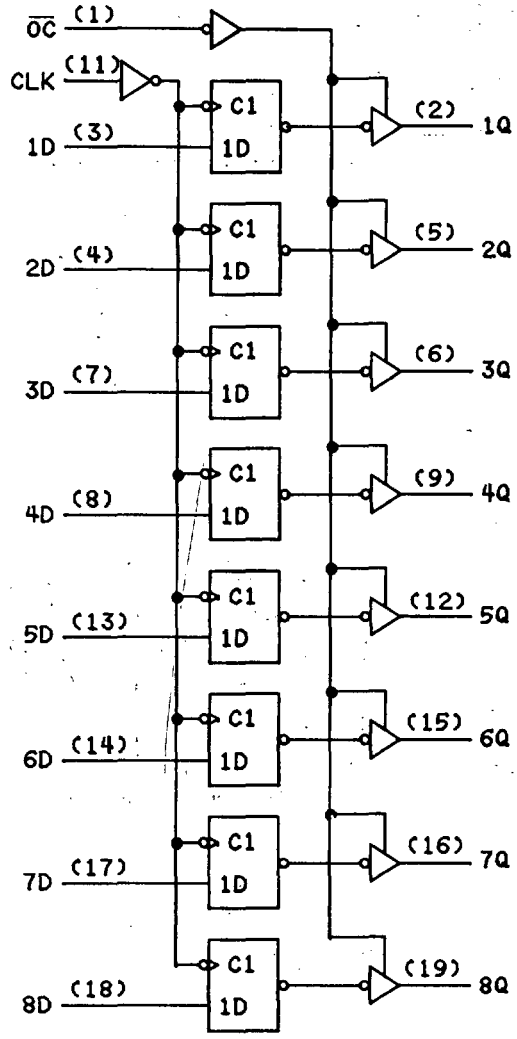


FIGURE 2. Logic diagram - Continued.

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Device type 01

INPUTS			OUTPUT
CLR	CLK	D	Q
L	X	X	L
H	↑	H	H
H	↑	L	L
H	L	X	Q_0

Device type 02

INPUTS			OUTPUT	
CLR	CLK	D	Q	$\overline{Q}$
L	X	X	L	H
H	↑	H	H	L
H	↑	L	L	H
H	L	X	Q_0	$\overline{Q}_0$

Device type 03

INPUTS			OUTPUT
$\overline{QC}$	ENABLE C	D	Q
L	H	H	H
L	H	L	L
L	L	X	Q_0
H	X	X	Z

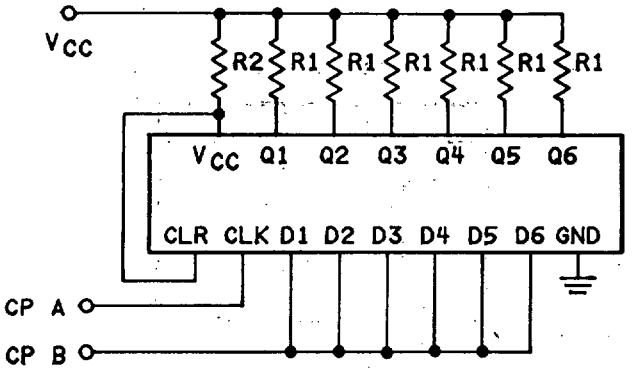
Device type 04

INPUTS			OUTPUT
$\overline{QC}$	CLK	D	D
L	↑	H	H
L	↑	L	L
L	L	X	Q_0
H	X	X	Z

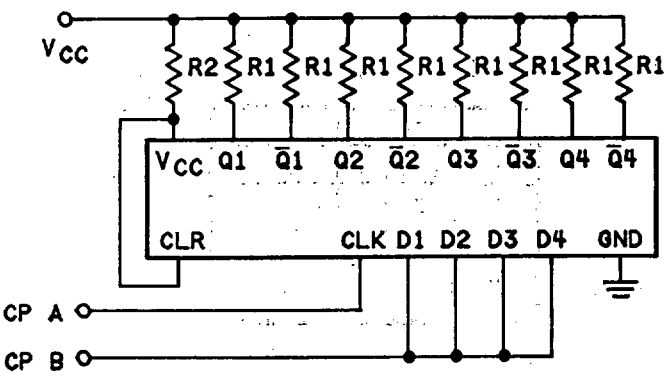
FIGURE 3. Truth tables.

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Device type 01

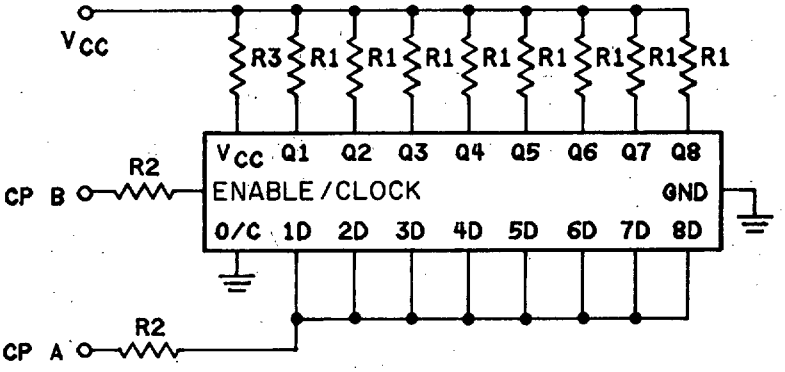


Device type 02



NOTES:
1. CP A = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to +0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
   CP B = Same as CP A, synchronized with CP A, except 50 kHz ±50% square wave.
2. R1 = 1 kΩ ±5%.
3. V_{CC} and R2 shall be chosen to insure 5.0 V minimum is present at device V_{CC} terminal.

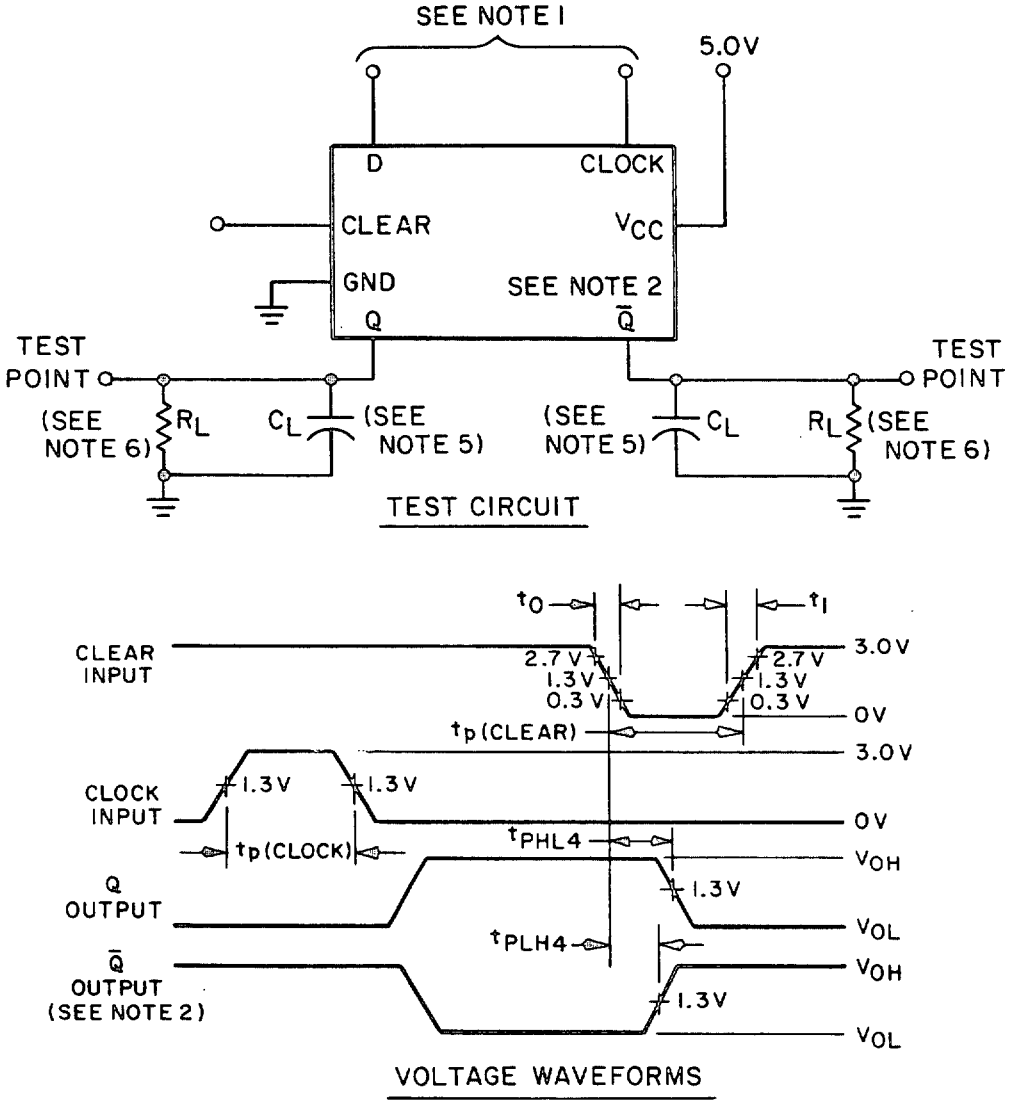
Device types 03 and 04



NOTES:
1. CP A = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to +0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
   CP B = Same as CP A, synchronized with CP A, except 50 kHz ±50% square wave.
2. R1 = 470Ω ±5%; R2 = 27Ω maximum.
3. V_{CC} and R3 shall be chosen to insure 5.0 V minimum is present at device V_{CC} terminal.

FIGURE 4. Burn-in and life test circuit.

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NOTES:
1. Clear input dominates regardless of the state of clock or D inputs.
2. $\overline{Q}$ output applies to device type 02 only.
3. Clear input pulse characteristics: $t_0 = t_1 = 6 \pm 1.5$ ns; $t_{p(clear)} = 15$ ns; PRR $\leq$ 1 MHz.
4. Inputs not under test are at ground.
5. $C_L = 50$ pF ±10%, including scope probe, wiring, and stray capacitance without package in test fixture.
6. $R_L = 499\Omega \pm 1\%$.
7. Clock input pulse characteristics: $t_{p(CLK)} \geq 12.5$ ns, PRR $\leq$ 1 MHz.
8. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 5. A synchronous switching test circuit for device types 01 and 02.

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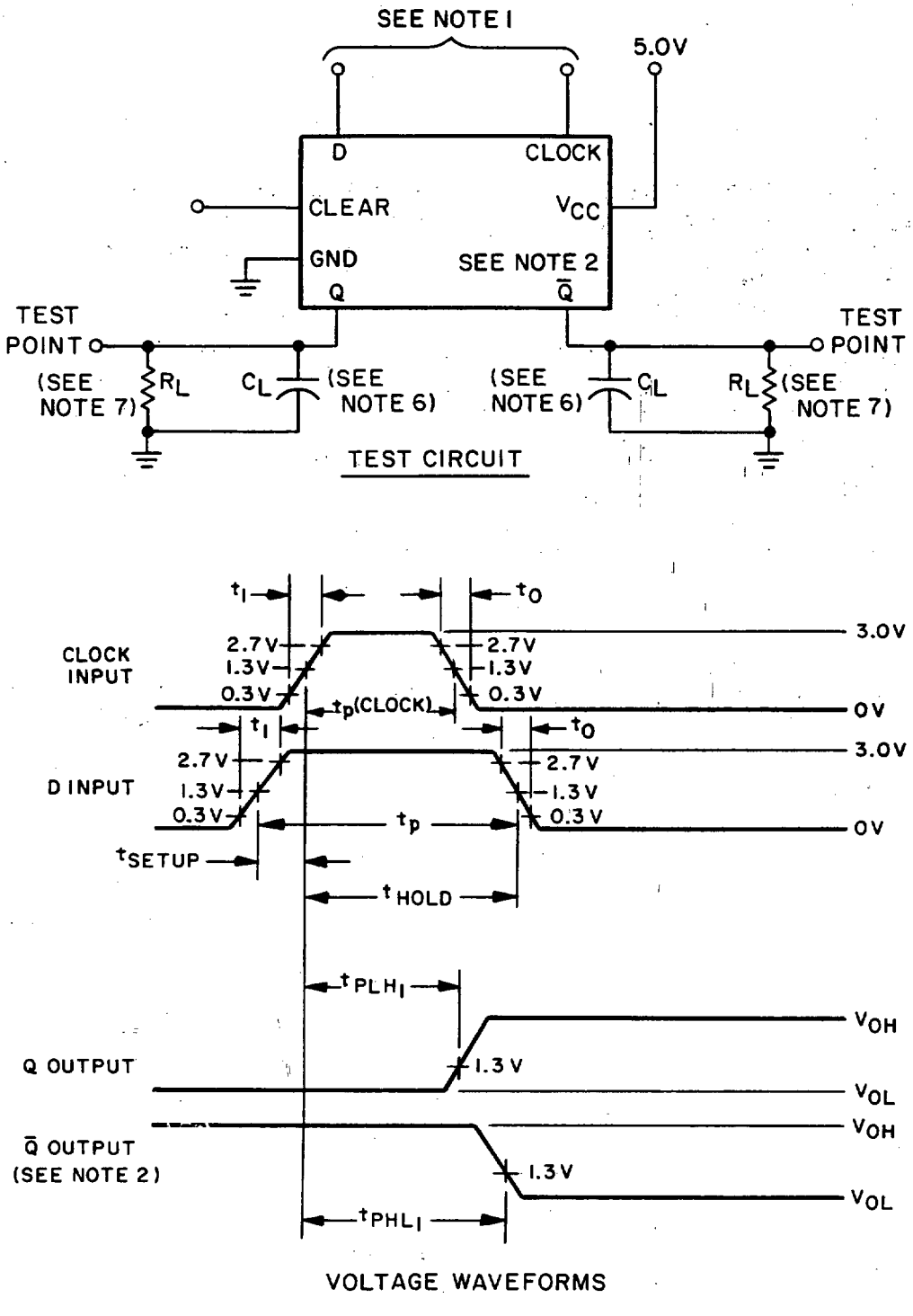


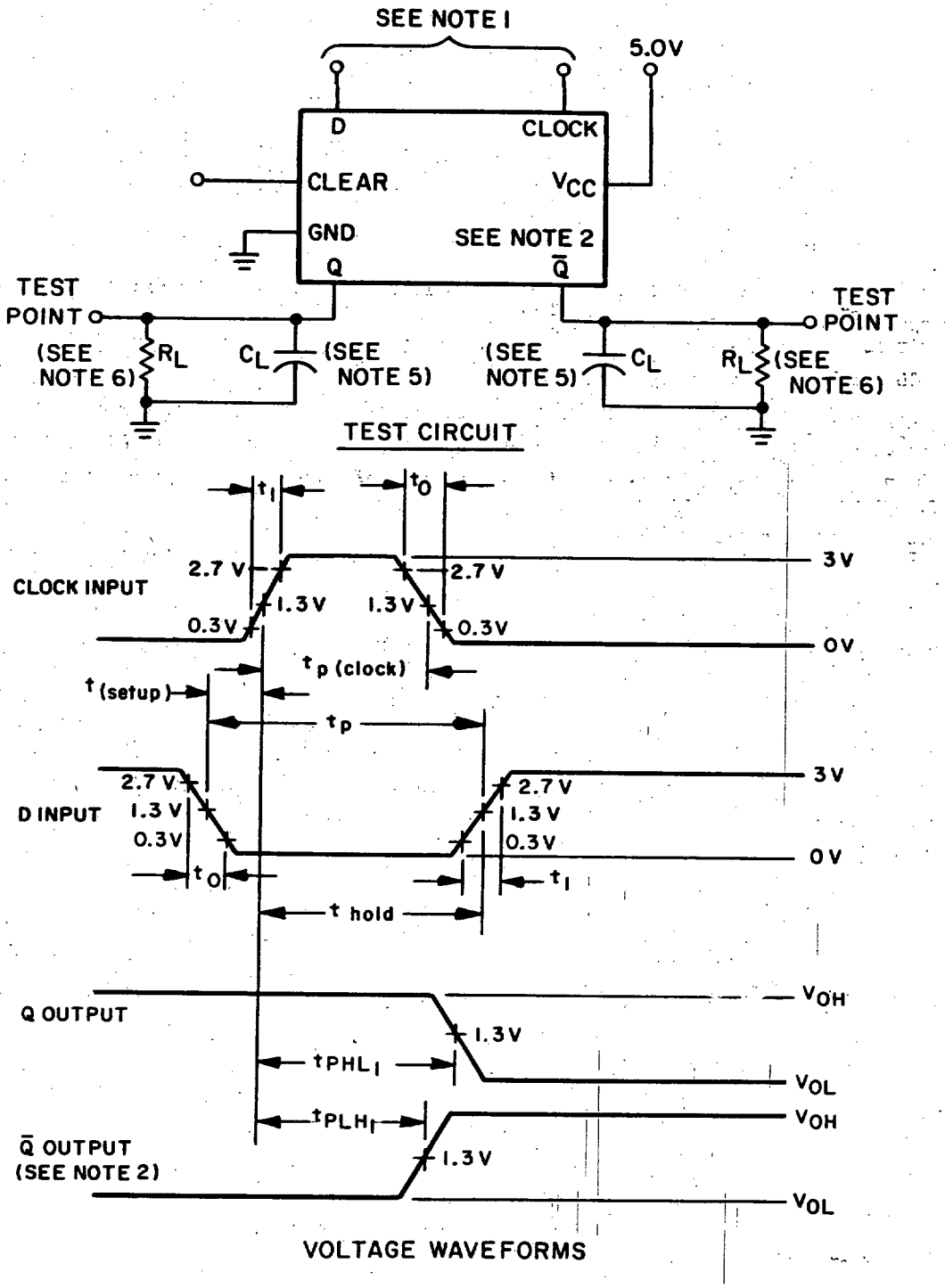
FIGURE 5. Synchronous switching test circuit (high-level data) types 01 and 02 - Continued.

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NOTES:
1. Clock input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; $t_{p(CLK)} = 12.5$ ns; PRR $\leq$ 1 MHz.
2. $\overline{Q}$ output applies to device type 02 only.
3. D input pulse characterisitics: $t_1 = t_0 = 6 \pm 1$ ns; $t_{SETUP} = 15$ ns; $t_{HOLD} = 0$ ns; $t_p = 15$ ns; PRR is 50% of clock PRR.
4. For f_{MAX}, the clock input pulse characteristics are as follows: $t_1 = t_0 \leq 3$ ns; for 25°C, $t_{p(CLK)} = 10$ ns; PRR = 50 MHz; for -55/125°C, $t_{p(CLK)} = 12.5$ ns, PRR = 40 MHz. The D input pulse shall be one-half of the frequency of the clock and the D + and + shall be coincident with the clock +. $t_1 = t_0 \leq 3$ ns.
5. Inputs not under test are at ground.
6. $C_L = 50$ pF ±10% including scope probe, wiring, and stray capacitance without package in test fixture.
7. $R_L = 499\Omega$ ±1%.
8. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 5. Synchronous switching test circuit (high-level data) types 01 and 02 - Continued.

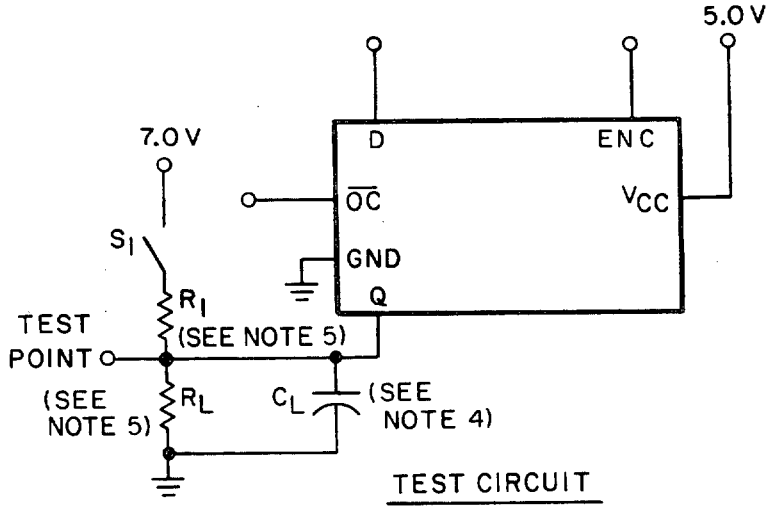
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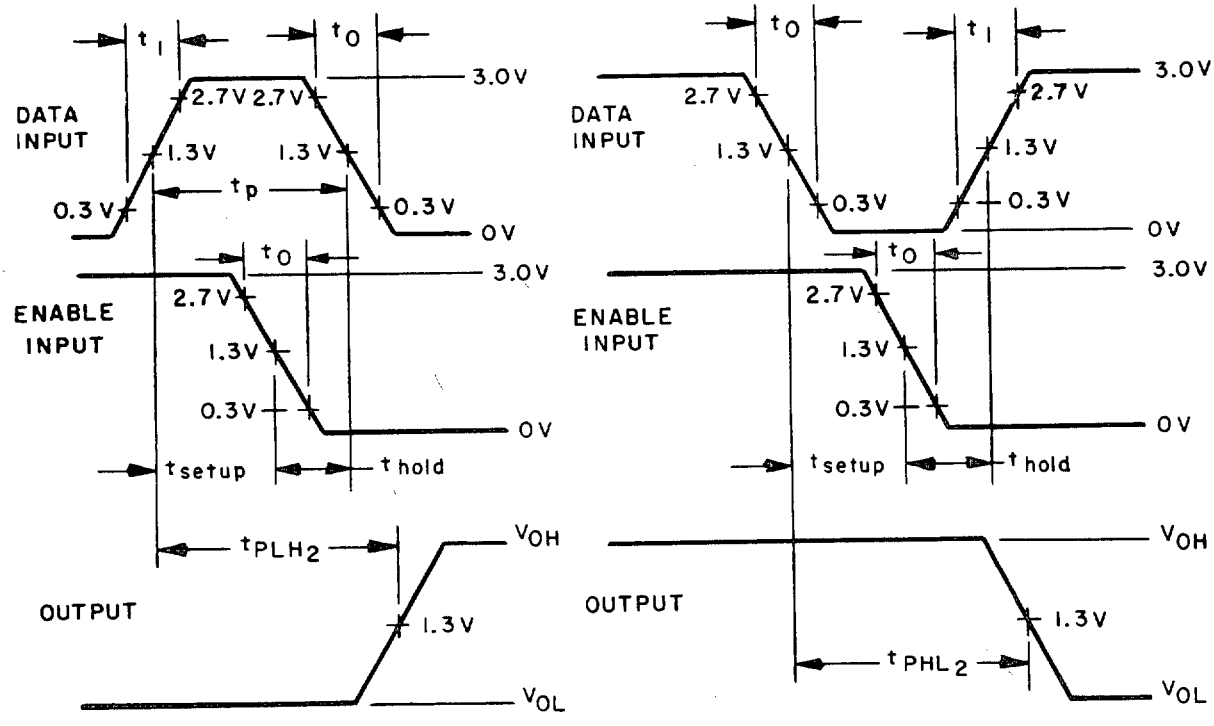
NOTES:
1. Clock input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; PRR $\leq$ 1 MHz.
2. $\overline{Q}$ output applies to device type 02 only.
3. D input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; $t_{SETUP} = 15$ ns; $t_{HOLD} = 0$ ns; PRR is 50% of clock PRR.
4. Inputs not under test are at ground.
5. C_L = 50 pF ±10%, including scope probe, wiring, and stray capacitance without package in test fixture.
6. $R_L = R1 = 499\Omega \pm 1\%$.
7. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 5. Synchronous switching test circuit (low-level data) types 01 and 02 - Continued.

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TEST CIRCUIT

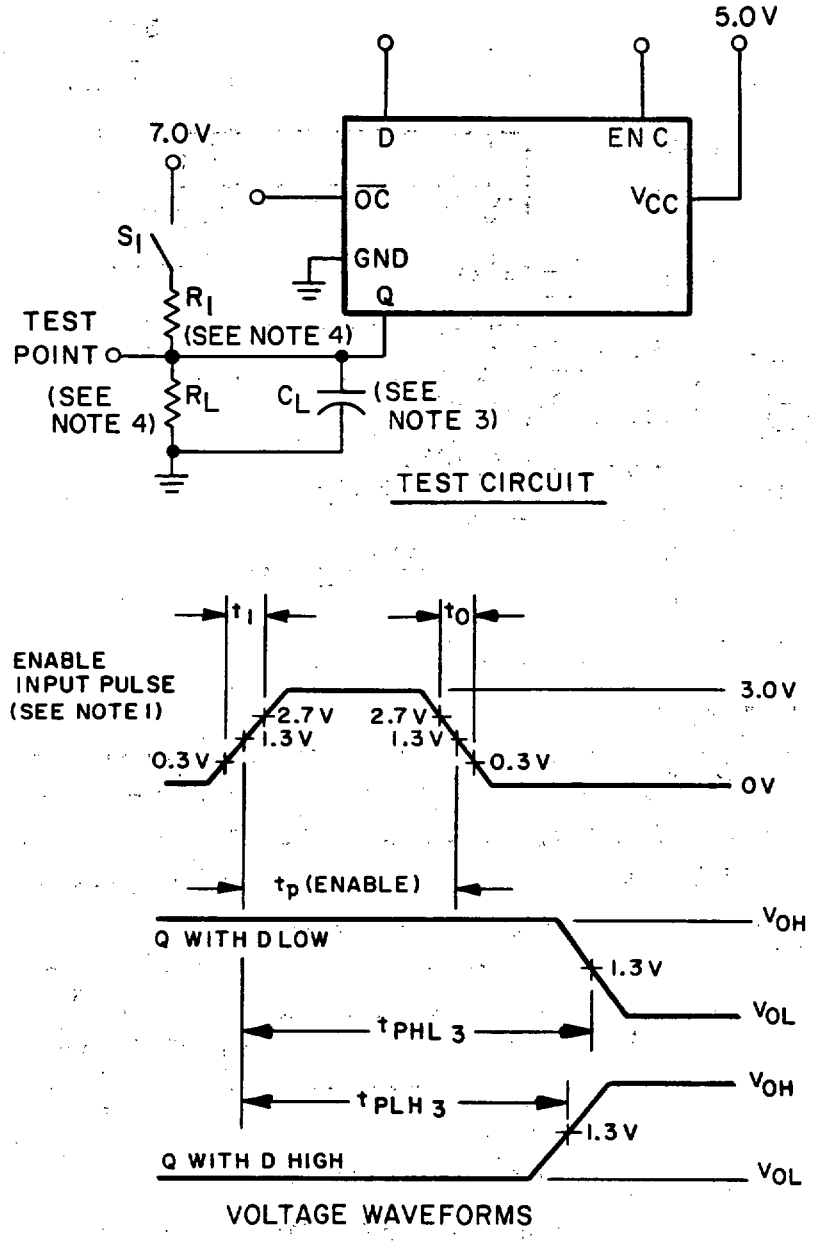


VOLTAGE WAVEFORMS

NOTES:
1. Enable input pulse characteristics: t_0 = 6 ±1.5 ns.
2. D input pulse characteristics: $t_1 = t_0$ = 6 ±1.5 ns; t_{SETUP} = 10 ns; t_{HOLD} = 7 ns; t_p = 17 ns.
3. Inputs not under test are at ground.
4. C_L = 50 pF ±10%, including scope probe, wiring, and stray capacitance without package in test fixture.
5. $R_L = R1$ = 499Ω ±1%.
6. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 6. Switching test circuit, type 03.

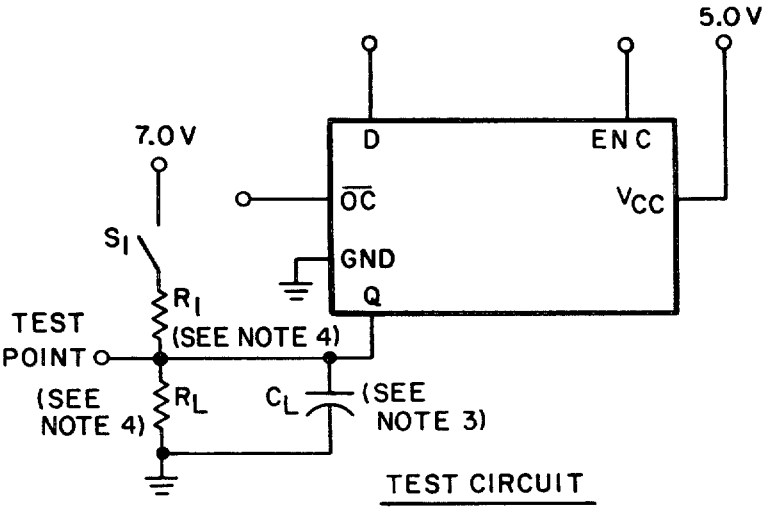
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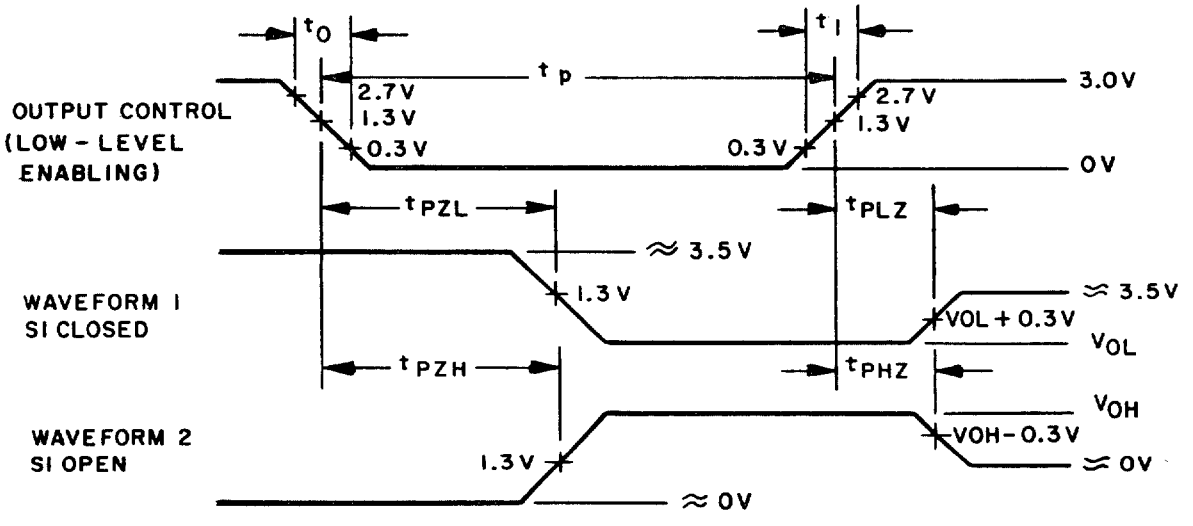
NOTES:
1. Enable input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; $t_p = 10$ ns; PRR $\leq$ 1 MHz.
2. Inputs not under test are at ground.
3. C_L = 50 pF ±10%, including scope probe, wiring, and stray capacitance without package in test fixture.
4. $R_L = R1 = 499\Omega \pm 1\%$.
5. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 6. Switching test circuit, type 03 - Continued.

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TEST CIRCUIT

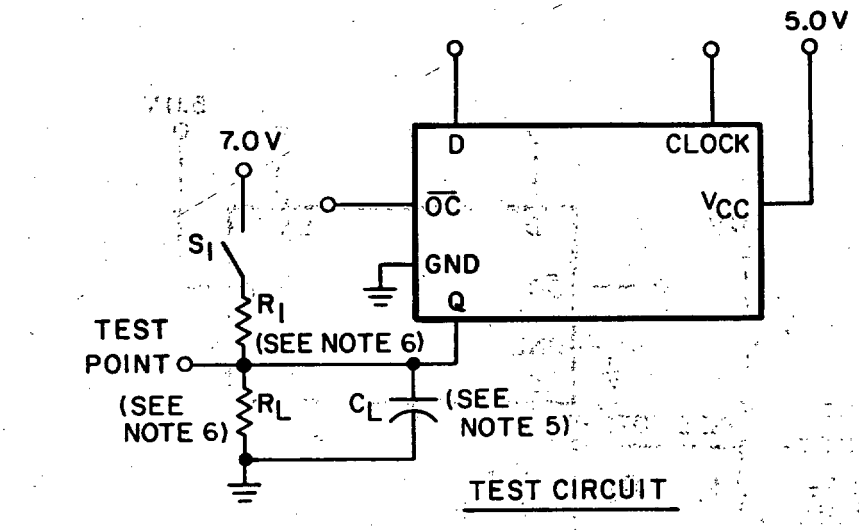


VOLTAGE WAVEFORMS

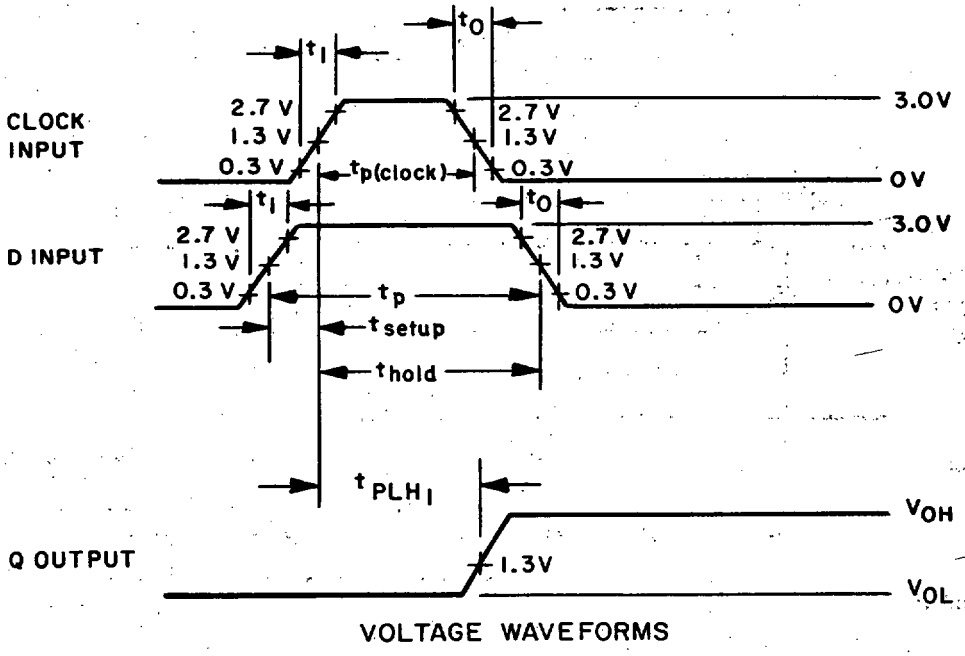
NOTES:
1. Output control input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; $t_p = 200$ ns; PRR $\leq$ 1 MHz.
2. Inputs not under test are at ground.
3. $C_L = 50$ pF ±10%, including scope probe, wiring, and stray capacitance without package in test fixture.
4. $R_L = R1 = 499\Omega \pm 1\%$.
5. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 6. Tri-state switching test circuit, type 03 - Continued.

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TEST CIRCUIT

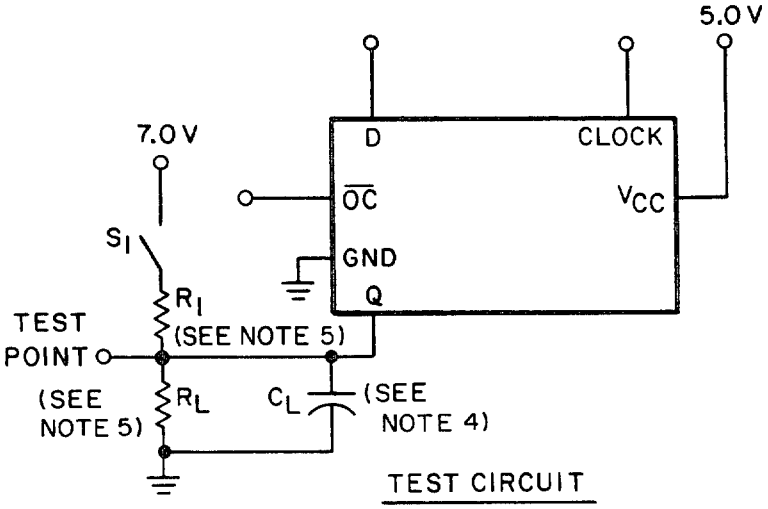


VOLTAGE WAVEFORMS

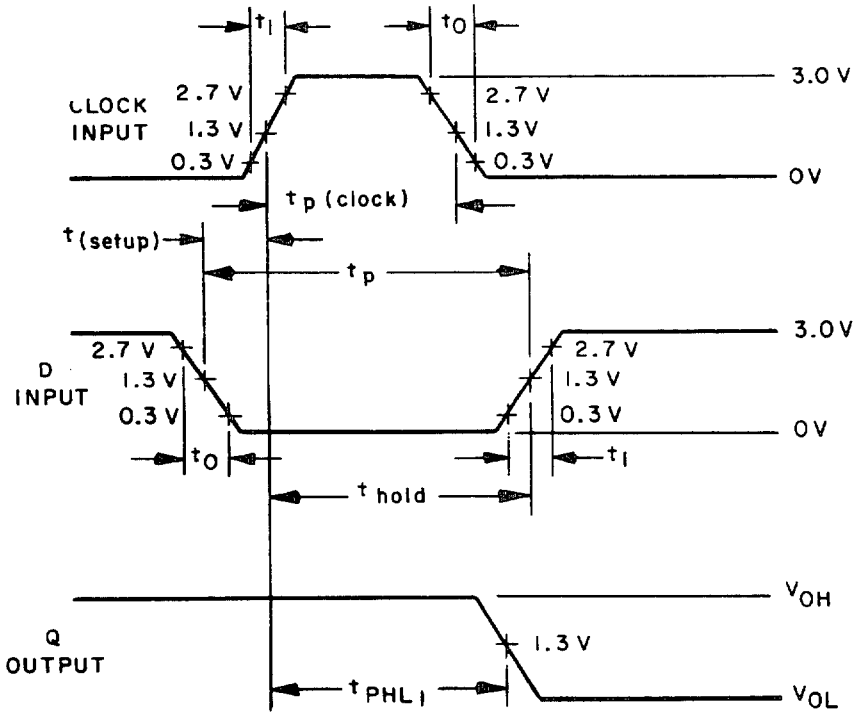
NOTES:
1. Clock input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; $t_{p(CLOCK)} = 16.5$ ns; PRR $\leq$ 1 MHz.
2. D input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; $t_{SETUP} = 10$ ns; $t_{HOLD} = 4$ ns; $t_p = 14$ ns; PRR is 50% of clock PRR.
3. For f_{MAX}, the clock input pulse are as follows: $t_1 = t_0 \leq 3$ ns; for 25°C, $t_{p(CLK)} = 14$ ns; PRR = 35 MHz; for -55/125°C, $t_{p(CLK)} = 16.5$ ns; PRR = 30 Mz; the D input pulse shall be one-half of the frequency of the clock and the D + and + shall be coincident with the clock +. $t_1 = t_0 \leq 3$ ns.
4. Inputs not under test are at ground.
5. $C_L = 50$ pF ±10%, including scope probe, wiring, and stray capacitance without package in test fixture.
6. $R_L = R1 = 499\Omega$ ±1%.
7. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 7. Synchronous switching test circuit (high-level data) type 04.

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TEST CIRCUIT

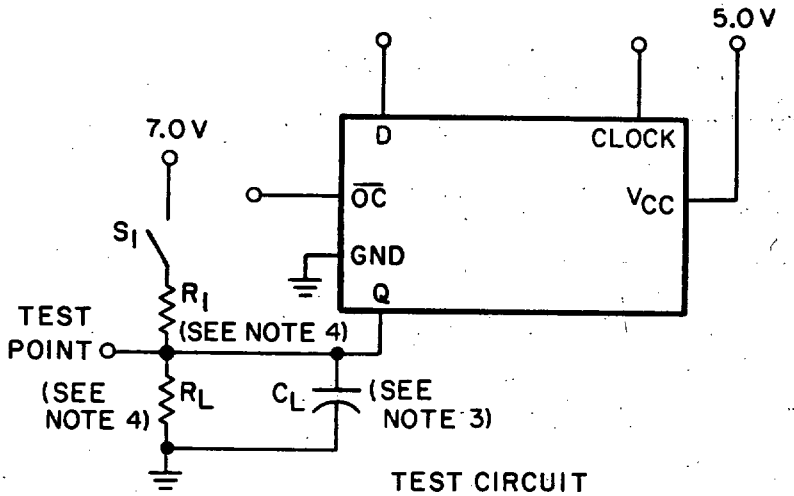


VOLTAGE WAVEFORMS

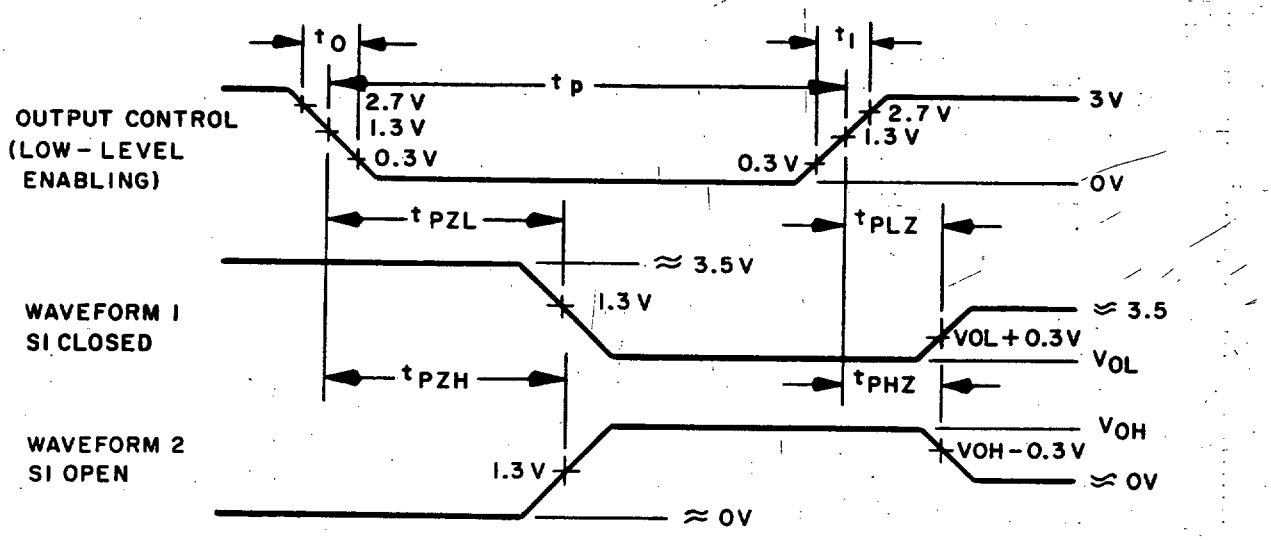
NOTES:
1. Clock input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; $t_{p(CLOCK)} = 12.5$ ns; PRR $\leq$ 1 MHz.
2. D input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; $t_{SETUP} = 10$ ns; $t_{HOLD} = 4$ ns; $t_p = 14$ ns; PRR is 50% of clock PRR.
3. Inputs not under test are at ground.
4. C_L = 50 pF ±10%, including scope probe, wiring, and stray capacitance without package in test fixture.
5. $R_L = R1 = 499\Omega \pm 1\%$.
6. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 7. Synchronous switching test circuit (low-level data) type 04 - Continued.

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TEST CIRCUIT



VOLTAGE WAVEFORMS

NOTES:
1. $\overline{OC}$ input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; $t_p \geq 200$ ns; PRR ≤ 1 MHz.
2. Inputs not under test are at ground.
3. $C_L = 50$ pF ±10%, including scope probe, wiring, and stray capacitance without package in test fixture.
4. $R_L = R1 = 499\Omega \pm 1\%$.
5. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 7. Tri-state switching test circuit, type 04 - Continued.

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Table III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Case 1/ 2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
				2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	$\overline{CLR}$	1Q	1D	2D	2Q	3D	3Q	GND	CLK	4Q	4D	5Q	5D	6D	6Q	V_{CC}				
1 T_C = +25°C	V_{OH}	3006	1	2.0 V	-400 µA	2.0 V					GND	2/							4.5 V	1Q	2.5		V
		"	2	"	"		2.0 V	-400 µA			"	"							"	2Q	"		"
		"	3	"	"				2.0 V	-400 µA	"	"							"	3Q	"		"
		"	4	"	"						"	"	-400 µA	2.0 V					"	4Q	"		"
		"	5	"	"						"	"			-400 µA	2.0 V			"	5Q	"		"
		"	6	"	"						"	"					2.0 V	-400 µA	"	6Q	"		"
	V_{OL}	3007	7	"	4 mA	0.8 V					"	"							"	1Q		0.4	"
		"	8	"	"		0.8 V	4 mA			"	"							"	2Q		"	"
		"	9	"	"				0.8 V	4 mA	"	"							"	3Q		"	"
		"	10	"	"						"	"	4 mA	0.8 V					"	4Q		"	"
		"	11	"	"						"	"			4 mA	0.8 V			"	5Q		"	"
		"	12	"	"						"	"					0.8 V	4 mA	"	6Q		"	"
	V_{IC}		13	-18 mA		-18 mA					"								"	$\overline{CLR}$		-1.5	"
			14				-18 mA				"								"	1D		"	"
			15						-18 mA		"								"	2D		"	"
			16								"	-18 mA							"	3D		"	"
			17								"								"	CLK		"	"
			18								"			-18 mA					"	4D		"	"
			19								"					-18 mA			"	5D		"	"
			20								"						-18 mA		"	6D		"	"
	I_{IL}	3009	21	0.4 V		0.4 V					"								5.5 V	$\overline{CLR}$	3/	3/	µA
		"	22				0.4 V				"								"	1D	"	"	"
		"	23						0.4 V		"								"	2D	"	"	"
		"	24								"	0.4 V							"	3D	"	"	"
		"	25								"			0.4 V					"	CLK	"	"	"
		"	26								"					0.4 V			"	4D	"	"	"
		"	27								"						0.4 V		"	5D	"	"	"
		"	28								"								"	6D	"	"	"
	I_{IH1}	3010	29	2.7 V		2.7 V					"								"			20	"
		"	30				2.7 V				"								"			"	"
		"	31						2.7 V		"								"			"	"
		"	32								"	2.7 V							"			"	"
		"	33								"			2.7 V					"			"	"
		"	34								"					2.7 V			"			"	"
		"	35								"						2.7 V		"			"	"
		"	36								"								"			"	"
	I_{IH2}	"	37	7.0 V		7.0 V					"								"			100	"
		"	38				7.0 V				"								"			"	"
		"	39						7.0 V		"								"			"	"
		"	40								"	7.0 V							"			"	"
		"	41								"			7.0 V					"			"	"
		"	42								"					7.0 V			"			"	"
		"	43								"						7.0 V		"			"	"
		"	44								"								"			"	"
	I_O 4/	3011	45	4.5 V	2.25 V	4.5 V					"	2/							"	1Q	-30	-112	mA
		"	46				4.5 V	2.25 V			"	"							"	2Q	"	"	"
		"	47						4.5 V	2.25 V	"	"							"	3Q	"	"	"
		"	48								"	"	2.25 V	4.5 V					"	4Q	"	"	"
		"	49								"	"			2.25 V	4.5 V			"	5Q	"	"	"
		"	50								"	"					4.5 V	2.25 V	"	6Q	"	"	"

See footnotes at end of device type 01.

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Table III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Case 1/ 2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
			Cases E, F Case 1/ 2 Test no.	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Cases E, F Case 1/ 2 Test no.	CLR	1Q	1D	2D	2Q	3D	3Q	GND	CLK	4Q	4D	5Q	5D	6D	6Q	V_{CC}				
1 T_C = +25°C	I_{CC}	3005	51	GND		GND	GND		GND		GND			GND		GND	GND		5.5 V	V_{CC}		19	mA
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and omit V_{IC} tests.																						
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and omit V_{IC} tests.																						
7 5/ T_C = +25°C	Truth table tests	3014	52 53 54 55 56	B 6/ A " " "	L H H L "	A 6/ A B "	A A B "	L H H L	A A B "	L H H L	GND " " "	B A B A B	L H H L L	A A B B B	L H H L L	A A B B B	A A B B B	L H H L L	4.5 V " " "	7/			
8	Same tests and terminal conditions as subgroup 7, except T_C = +125°C and -55°C.																						
9 T_C = +25°C	f_{MAX} 8/	3003 (fig 5)	57 58 59 60 61 62	3.0 V " " " " "	OUT	IN	IN	OUT	IN	OUT	GND " " " " "	IN " " " " "	OUT	IN	OUT	IN	IN	OUT	5.0 V " " " " "	1Q 2Q 3Q 4Q 5Q 6Q		50 " " " " "	MHz " " " " "
	t_{PLH1}	"	63 64 65 66 67 68	" " " " " "	OUT	IN	IN	OUT	IN	OUT	" " " " " "	" " " " " "	OUT	IN	OUT	IN	IN	OUT	" " " " " "	CLK > 1Q CLK > 2Q CLK > 3Q CLK > 4Q CLK > 5Q CLK > 6Q	3 " " " " "	15 " " " " "	ns " " " " "
	t_{PHL1}	"	69 70 71 72 73 74	" " " " " "	OUT	IN	IN	OUT	IN	OUT	" " " " " "	" " " " " "	OUT	IN	OUT	IN	IN	OUT	" " " " " "		5 " " " " "	17 " " " " "	" " " " " "
	t_{PHL4}	"	75 76 77 78 79 80	IN " " " " "	OUT	3.0 V	3.0 V	OUT	3.0 V	OUT	" " " " " "	2/ " " " " "	OUT	3.0 V	OUT	3.0 V	3.0 V	OUT	" " " " " "		8 " " " " "	23 " " " " "	" " " " " "
10 T_C =+125°C	f_{MAX}	Same tests and terminal conditions as subgroup 9, except T_C = +125°C.																				40	MHz
	t_{PLH1}																				3	17	ns
	t_{PHL1}																				5	20	"
	t_{PHL4}																				8	26	"
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																						

NOTES:

1/ Terminals not referenced are N/C.

2/ Apply 2.0 V min/4.5 V max, 0 V, (pulse, prior to test).

3/ I_{IL} limits shall be as follows:

Test	Min/Max limits in µA for circuits A	B	C
I_{IL}	0/-100		

4/ Method 3011 of MIL-STD-883 shall be used, except the output shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one-half of the true short-circuit output current I_{OS}.

5/ Tests shall be performed in sequence, attributes data only.

6/ A = 3.0 V minimum, B = 0.0 V or GND.

7/ H > 1.5 V, L < 1.5 V.

8/ F_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

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Table III. Group A inspection for device type 02.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Case 1/ 2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
				2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	$\overline{CLR}$	1Q	$1\overline{Q}$	1D	2D	$2\overline{Q}$	2Q	GND	CLK	3Q	$3\overline{Q}$	3D	4D	$4\overline{Q}$	4Q	V_{CC}				
1 T_C = +25°C	V_{OH}	3006	1	0.8 V		-400 µA			-400 µA		GND			-400 µA			-400 µA		4.5 V	$1\overline{Q}$	2.5		V
		"	2	"		"			"		"			"			"		"	$2\overline{Q}$	"		"
		"	3	"		"			"		"			"			"		"	$3\overline{Q}$	"		"
		"	4	"		"			"		"			"			"		"	$4\overline{Q}$	"		"
		"	5	2.0 V	-400 µA		2.0 V	2.0 V		-400 µA	"	2/	-400 µA		2.0 V				"	1Q	"		"
		"	6	"	"		"			"	"	"			"				"	2Q	"		"
		"	7	"	"		"			"	"	"			"				"	3Q	"		"
		"	8	"	"		"			"	"	"			"				"	4Q	"		"
		"	9	"	"	-400 µA	0.8 V		-400 µA		"	"			"				"	$1\overline{Q}$	"		"
		"	10	"	"			0.8 V			"	"			"				"	$2\overline{Q}$	"		"
		"	11	"	"						"	"		-400 µA	0.8 V				"	$3\overline{Q}$	"		"
		"	12	"	"						"	"			"				"	$4\overline{Q}$	"		"
	V_{OL}	3007	13	0.8 V	4 mA					4 mA	"								"	1Q		0.4	"
		"	14	"							"		4 mA					4 mA	"	2Q		"	"
		"	15	"							"								"	3Q		"	"
		"	16	"							"								"	4Q		"	"
		"	17	2.0 V	"	4 mA	2.0 V	2.0 V	4 mA	"	"	2/	"	4 mA	2.0 V	2.0 V	4 mA	"	"	$1\overline{Q}$		"	"
		"	18	"	"						"								"	$2\overline{Q}$		"	"
		"	19	"	"						"								"	$3\overline{Q}$		"	"
		"	20	"	"						"								"	$4\overline{Q}$		"	"
		"	21	"	4 mA		0.8 V			4 mA	"								"	1Q		"	"
		"	22	"							"								"	2Q		"	"
		"	23	"							"								"	3Q		"	"
		"	24	"							"								"	4Q		"	"
	V_{IC}		25	-18 mA							"								"	$\overline{CLR}$		-1.5	"
			26				-18 mA	-18 mA			"								"	1D		"	"
			27								"								"	2D		"	"
			28								"	-18 mA							"	CLK		"	"
			29								"				-18 mA				"	3D		"	"
			30								"					-18 mA			"	4D		"	"
	I_{IL}	3009	31	0.4 V							"								5.5 V	$\overline{CLR}$	3/	3/	µA
		"	32				0.4 V				"								"	1D	"	"	"
		"	33					0.4 V			"								"	2D	"	"	"
		"	34								"	0.4 V							"	CLK	"	"	"
		"	35								"				0.4 V				"	3D	"	"	"
		"	36								"					0.4 V			"	4D	"	"	"
	I_{IH1}	3010	37	2.7 V							"								"	$\overline{CLR}$		20	"
		"	38				2.7 V				"								"	1D		"	"
		"	39					2.7 V			"								"	2D		"	"
		"	40								"	2.7 V							"	CLK		"	"
		"	41								"				2.7 V				"	3D		"	"
		"	42								"					2.7 V			"	4D		"	"
	I_{IH2}	"	43	7.0 V							"								"	$\overline{CLR}$		100	"
		"	44				7.0 V				"								"	1D		"	"
		"	45					7.0 V			"								"	2D		"	"
		"	46								"	7.0 V							"	CLK		"	"
		"	47								"				7.0 V				"	3D		"	"
		"	48								"					7.0 V			"	4D		"	"
	I_O 4/	3011	49	4.5 V	2.25 V	2.25 V	4.5 V GND				"								"	$\overline{CLR}$	-30	-112	mA
		"	50	"				4.5 V GND	2.25 V		"								"	1Q	"	"	"
		"	51	"						2.25 V	"								"	$1\overline{Q}$	"	"	"
		"	52	"							"								"	1D	"	"	"
		"	53	"							"								"	2D	"	"	"
		"	54	"							"								"	$2\overline{Q}$	"	"	"
		"	55	"							"								"	CLK	"	"	"
		"	56	"							"								"	$3\overline{Q}$	"	"	"
		"		"							"								"	3D	"	"	"
		"		"							"								"	$4\overline{Q}$	"	"	"
		"		"							"								"	4D	"	"	"

See footnotes at end of device type 02.

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Table III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Case 1/ 2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
				CLR	1Q	1$\overline{Q}$	1D	2D	2$\overline{Q}$	2Q	GND	CLK	3Q	3$\overline{Q}$	3D	4D	4$\overline{Q}$	4Q	V_{CC}				
1 T_C = +25°C	I_{CC}	3005	57	GND			GND	GND			GND	2/			GND	GND			5.5 V	V_{CC}		14	mA
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and omit V_{IC} tests.																						
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and omit V_{IC} tests.																						
7 T_C = +25°C 5/ 6/	Truth table tests	3014	58	B	L	H	A	A	H	L	GND	B	L	H	A	A	H	L	4.5 V	7/			
	"	"	59	A	H	L	A	A	L	H	"	A	H	L	A	A	L	H	"				
	"	"	60	"	H	L	B	B	L	H	"	B	H	L	B	B	L	H	"				
	"	"	61	"	L	H	B	B	H	L	"	A	L	H	B	B	H	L	"				
	"	"	62	"	L	H	A	A	H	L	"	B	L	H	A	A	H	L	"				
	"	"	63	"	H	L	"	"	L	H	"	A	H	L	A	A	L	H	"				
	"	"	64	B	L	H	"	"	H	L	"	B	L	H	A	A	H	L	"				
	"	"	65	B	L	H	"	"	H	L	"	A	L	H	A	A	H	L	"				
	"	"	66	B	L	H	"	"	H	L	"	B	L	H	A	A	H	L	"				
8	Same tests and terminal conditions as subgroup 7, except T_C = +125°C and -55°C.																						
9 T_C = +25°C	f_{MAX} 8/	3003 (fig 5)	67	3.0 V	OUT	OUT	IN			OUT	GND	IN	OUT	OUT	IN	IN	OUT	OUT	5.0 V	1Q		50	MHz
	"	"	68	"			IN				"	"			IN				"	1$\overline{Q}$		"	"
	"	"	69	"				IN			"	"							"	2Q		"	"
	"	"	70	"				IN	OUT		"	"							"	2$\overline{Q}$		"	"
	"	"	71	"							"	"							"	3Q		"	"
	"	"	72	"							"	"							"	3$\overline{Q}$		"	"
	"	"	73	"							"	"							"	4Q		"	"
	"	"	74	"							"	"							"	4$\overline{Q}$		"	"
	t_{PLH1}	"	75	"	OUT		IN			OUT	"	"	OUT		IN				"	CLK > 1$\overline{Q}$	3	15	ns
	"	"	76	"				IN			"	"							"	CLK > 2Q	"	"	"
	"	"	77	"							"	"							"	CLK > 3Q	"	"	"
	"	"	78	"							"	"				IN			"	CLK > 4Q	"	"	"
	"	"	79	"		OUT	IN				"	"						OUT	"	CLK > 1$\overline{Q}$	"	"	"
	"	"	80	"				IN	OUT		"	"							"	CLK > 2$\overline{Q}$	"	"	"
	"	"	81	"							"	"		OUT	IN				"	CLK > 3$\overline{Q}$	"	"	"
	"	"	82	"							"	"				IN	OUT		"	CLK > 4Q	"	"	"
	t_{PHL1}	"	83	"	OUT		IN			OUT	"	"	OUT		IN				"	CLK > 1Q	5	17	"
	"	"	84	"				IN			"	"							"	CLK > 2Q	"	"	"
	"	"	85	"							"	"							"	CLK > 3Q	"	"	"
	"	"	86	"							"	"				IN			"	CLK > 4Q	"	"	"
	"	"	87	"		OUT	IN				"	"						OUT	"	CLK > 1$\overline{Q}$	"	"	"
	"	"	88	"				IN	OUT		"	"							"	CLK > 2$\overline{Q}$	"	"	"
	"	"	89	"							"	"		OUT	IN				"	CLK > 3$\overline{Q}$	"	"	"
	"	"	90	"							"	"				IN	OUT		"	CLK > 4Q	"	"	"
	t_{PLH4}	"	91	IN		OUT	3.0 V				"	2/							"	CLR > 1$\overline{Q}$	5	18	"
	"	"	92					3.0 V	OUT		"	"							"	CLR > 2Q	"	"	"
	"	"	93								"	"		OUT	3.0 V				"	CLR > 3$\overline{Q}$	"	"	"
	"	"	94								"	"				3.0 V	OUT		"	CLR > 4Q	"	"	"
	t_{PHL4}	"	95	IN	OUT		3.0 V				"	"							"	CLR > 1Q	8	23	"
	"	"	96					3.0 V		OUT	"	"							"	CLR > 2Q	"	"	"
	"	"	97								"	"	OUT		3.0 V				"	CLR > 3Q	"	"	"
	"	"	98								"	"				3.0 V		OUT	"	CLR > 4Q	"	"	"

See footnotes at end of device type 02.

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Table III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	$\overline{CLR}$	1Q	$1\overline{Q}$	1D	2D	$2\overline{Q}$	2Q	GND	CLK	3Q	$3\overline{Q}$	3D	4D	$4\overline{Q}$	4Q	V_{CC}				
10 T_C =+125°C	f_{MAX}	Same tests and terminal conditions as subgroup 9, except T_C = +125°C.																				40	MHz
	t_{PLH1}																				3	17	ns
	t_{PHL1}																				5	20	"
	t_{PLH4}																				5	20	"
	t_{PHL4}																				8	26	"
11 T_C = -55°C	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																						

NOTES:

1/ Terminals not referenced are N/C.

2/ Apply  2.0 V min/4.5 V max, 0 V (pulse, prior to test).

3/ I_{IL} limits shall be as follows:

Test	Min/Max limits in µA for circuits		
	A	B	C
I_{IL}	0/-100		

4/ Method 3011 of MIL-STD-883 shall be used, except the output shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one-half of the true short-circuit output current I_{OS}.

5/ Tests shall be performed in sequence, attributes data only.

6/ A = 3.0 V minimum, B = 0.0 V or GND.

7/ H > 1.5 V, L < 1.5 V.

8/ F_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

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TABLE III. Group A inspection for device type 03.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{OC}$	1Q	1D	2D	2Q	3Q	3D	4D	4Q	GND	ENG	5Q	5D	6D	6Q	7Q	7D	8D	8Q	V_{CC}				
1 T_C = +25°C	V_{OH}	3006	1	0.8 V	-1 mA	2.0 V								2.0 V									4.5 V	1Q	2.4		V
			2	"			2.0 V	-1 mA						"									"	2Q	"		"
			3	"					-1 mA	2.0 V				"									"	3Q	"		"
			4	"							2.0 V	-1 mA		"									"	4Q	"		"
			5	"										"	-1 mA	2.0 V							"	5Q	"		"
			6	"										"			2.0 V	-1 mA					"	6Q	"		"
			7	"										"					-1 mA	2.0 V			"	7Q	"		"
			8	"										"							2.0 V	-1 mA	"	8Q	"		"
	V_{OL}	3007	9	"	12 mA	0.8 V								"									"	1Q		0.4	"
			10	"			0.8 V	12 mA						"									"	2Q		"	"
			11	"					12 mA	0.8 V				"									"	3Q		"	"
			12	"							0.8 V	12 mA		"									"	4Q		"	"
			13	"										"	12 mA	0.8 V							"	5Q		"	"
			14	"										"			0.8 V	12 mA					"	6Q		"	"
			15	"										"					12 mA	0.8 V			"	7Q		"	"
			16	"										"							0.8 V	12 mA	"	8Q		"	"
	V_{IC}		17	-18 mA										"									"	$\overline{OC}$		-1.5	"
			18			-18 mA								"									"	1D		"	"
			19				-18 mA							"									"	2D		"	"
			20							-18 mA				"									"	3D		"	"
			21								-18 mA			"									"	4D		"	"
			22											"									"	ENG		"	"
			23											-18 mA									"	5D		"	"
			24											"		-18 mA							"	6D		"	"
			25											"			-18 mA						"	7D		"	"
			26											"						-18 mA	-18 mA		"	8D		"	"
	I_{IL}	3009	27	0.4 V										"									5.5 V	$\overline{OC}$	1/	1/	μA
			28			0.4 V								"									"	1D	"	"	"
			29				0.4 V							"									"	2D	"	"	"
			30							0.4 V				"									"	3D	"	"	"
			31								0.4 V			"									"	4D	"	"	"
			32											"									"	ENG	"	"	"
			33											0.4 V									"	5D	"	"	"
			34											"		0.4 V							"	6D	"	"	"
			35											"			0.4 V						"	7D	"	"	"
			36											"						0.4 V	0.4 V		"	8D	"	"	"
	I_{IH1}	3010	37	2.7 V										"									5.5 V	$\overline{OC}$		20	"
			38			2.7 V								"									"	1D		"	"
			39				2.7 V							"									"	2D		"	"
			40							2.7 V				"									"	3D		"	"
			41								2.7 V			"									"	4D		"	"
			42											"									"	ENG		"	"
			43											2.7 V									"	5D		"	"
			44											"		2.7 V							"	6D		"	"
			45											"			2.7 V						"	7D		"	"
			46											"						2.7 V	2.7 V		"	8D		"	"
	I_{IH2}	"	47	7.0 V										"									"	$\overline{OC}$		100	"
			48			7.0 V								"									"	1D		"	"
			49				7.0 V							"									"	2D		"	"
			50							7.0 V				"									"	3D		"	"
			51								7.0 V			"									"	4D		"	"
			52											"									"	ENG		"	"
			53											7.0 V									"	5D		"	"
			54											"		7.0 V							"	6D		"	"
			55											"			7.0 V						"	7D		"	"
			56											"						7.0 V	7.0 V		"	8D		"	"
	I_{OZL}		57	2.0 V	0.4 V	2.0 V								"									"	1Q		-20	"
			58	"			2.0 V	0.4 V						"									"	2Q		"	"
			59	"					0.4 V	2.0 V				"									"	3Q		"	"
			60	"							2.0 V	0.4 V		"									"	4Q		"	"
			61	"										"	0.4 V	2.0 V							"	5Q		"	"
			62	"										"			2.0 V	0.4 V					"	6Q		"	"
			63	"										"					0.4 V	2.0 V			"	7Q		"	"
			64	"										"							2.0 V	0.4 V	"	8Q		"	"

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				OC	1Q	1D	2D	2Q	3Q	3D	4D	4Q	GND	ENG	5Q	5D	6D	6Q	7Q	7D	8D	8Q	V_{CC}				
1 T_C = +25°C	I_{OZH}		65	2.0 V	2.7 V	0.8 V							GND										5.5 V	1Q		20	µA
			66	"			0.8 V	2.7 V					"										"	2Q		"	"
			67	"					2.7 V	0.8 V			"										"	3Q		"	"
			68	"							0.8 V	2.7 V	"		2.7 V	0.8 V							"	4Q		"	"
			69	"									"				0.8 V	2.7 V					"	5Q		"	"
			70	"									"						2.7 V	0.8 V			"	6Q		"	"
			71	"									"								0.8 V	2.7 V	"	7Q		"	"
			72	"									"										"	8Q		"	"
	I_{O} 2/	3011	73	GND	2.25 V	4.5 V							"	4.5 V									"	1Q	-30	-112	mA
		"	74				4.5 V	2.25 V					"	"									"	2Q	"	"	"
		"	75						2.25 V	4.5 V			"	"									"	3Q	"	"	"
		"	76								4.5 V	2.25 V	"	"	2.25 V	4.5 V							"	4Q	"	"	"
		"	77	GND									"	"			4.5 V	2.25 V					"	5Q	"	"	"
		"	78	"									"	"					2.25 V	4.5 V			"	6Q	"	"	"
		"	79	"									"	"							4.5 V	2.25 V	"	7Q	"	"	"
		"	80	"									"	"									"	8Q	"	"	"
	I_{CCH}	3005	81	"		4.5 V	4.5 V			4.5 V	4.5 V		"	"		4.5 V	4.5 V			4.5 V	4.5 V		"	V_{CC}		16	"
	I_{CCL}	"	82	"		GND	GND			GND	GND		"	"		GND	GND			GND	GND		"	V_{CC}		25	"
	I_{CCZ}	"	83	4.5 V		4.5 V	4.5 V			4.5 V	4.5 V		"			4.5 V	4.5 V			4.5 V	4.5 V		"	V_{CC}		27	"
2	Same tests, terminal conditions, and tests as subgroup 1, except T_C = +125°C and omit V_{IC} tests.																										
3	Same tests, terminal conditions, and tests as subgroup 1, except T_C = -55°C and omit V_{IC} tests.																										
7 T_C = +25°C	Truth table tests 3/ 4/	3014	84	B	H	A	A	H	H	A	A	H	GND	A	H	A	A	H	H	A	A	H	4.5 V	5/			
		"	85	"	L	B	B	L	L	B	B	L	"	A	L	B	B	L	L	B	B	L	"				
		"	86	"	L	B	A	L	L	A	A	L	"	B	L	A	A	L	L	A	A	L	"				
		"	87	"	L	A	A	H	H	A	A	H	"	B	H	A	A	H	H	A	A	H	"				
		"	88	"	H	A	A	H	H	A	A	H	"	A	H	B	A	H	H	A	A	H	"				
		"	89	"	H	A	A	H	H	A	A	H	"	B	H	B	A	H	H	A	A	H	"				
		"	90	"	H	B	B	H	H	B	B	H	"	B	H	B	B	H	H	B	B	H	"				
8	Same tests and terminal conditions as subgroup 7, except T_C = +125°C and -55°C.																										
9 T_C = +25°C	t_{PLH2}	3003 (fig. 6)	91	GND	OUT	IN	IN	OUT	OUT	IN	IN	OUT	GND	IN	OUT	IN	IN	OUT	OUT	IN	IN	OUT	5.0 V	1D > 1Q	2	12	ns
		"	92	"									"	"									"	2D > 2Q	"	"	"
		"	93	"									"	"									"	3D > 3Q	"	"	"
		"	94	"									"	"									"	4D > 4Q	"	"	"
		"	95	"									"	"									"	5D > 5Q	"	"	"
		"	96	"									"	"									"	6D > 6Q	"	"	"
		"	97	"									"	"									"	7D > 7Q	"	"	"
		"	98	"									"	"									"	8D > 8Q	"	"	"
	t_{PHL2}	"	99	"	OUT	IN	IN	OUT	OUT	IN	IN	OUT	"	"	OUT	IN	IN	OUT	OUT	IN	IN	OUT	"	1D > 1Q	4	16	"
		"	100	"									"	"									"	2D > 2Q	"	"	"
		"	101	"									"	"									"	3D > 3Q	"	"	"
		"	102	"									"	"									"	4D > 4Q	"	"	"
		"	103	"									"	"									"	5D > 5Q	"	"	"
		"	104	"									"	"									"	6D > 6Q	"	"	"
		"	105	"									"	"									"	7D > 7Q	"	"	"
		"	106	"									"	"									"	8D > 8Q	"	"	"
	t_{PLH3}	"	107	"	OUT	3.0 V	3.0 V	OUT	OUT	3.0 V	3.0 V	OUT	"	"	OUT	3.0 V	3.0 V	OUT	OUT	3.0 V	3.0 V	OUT	"	ENG > 1Q	6	22	"
		"	108	"									"	"									"	ENG > 2Q	"	"	"
		"	109	"									"	"									"	ENG > 3Q	"	"	"
		"	110	"									"	"									"	ENG > 4Q	"	"	"
		"	111	"									"	"									"	ENG > 5Q	"	"	"
		"	112	"									"	"									"	ENG > 6Q	"	"	"
		"	113	"									"	"									"	ENG > 7Q	"	"	"
		"	114	"									"	"									"	ENG > 8Q	"	"	"
	t_{PHL3}	"	115	"	OUT	GND	GND	OUT	OUT	GND	GND	OUT	"	"	OUT	GND	GND	OUT	OUT	GND	GND	OUT	"	ENG > 1Q	7	23	"
		"	116	"									"	"									"	ENG > 2Q	"	"	"
		"	117	"									"	"									"	ENG > 3Q	"	"	"
		"	118	"									"	"									"	ENG > 4Q	"	"	"
		"	119	"									"	"									"	ENG > 5Q	"	"	"
		"	120	"									"	"									"	ENG > 6Q	"	"	"
		"	121	"									"	"									"	ENG > 7Q	"	"	"
		"	122	"									"	"									"	ENG > 8Q	"	"	"

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{OE}$	1Q	1D	2D	2Q	3Q	3D	4D	4Q	GND	ENG	5Q	5D	6D	6Q	7Q	7D	8D	8Q	V_{CC}				
9 T_C = +25°C	t_{PZH}	3003 (fig. 6)	123	IN	OUT	3.0 V	3.0 V	OUT	OUT	3.0 V	3.0 V	OUT	GND		OUT	3.0 V	3.0 V	OUT	OUT	3.0 V	3.0 V	OUT	5.0 V	$\overline{OE}$ to 1Q	5	20	ns
			124	"									"										"	$\overline{OE}$ to 2Q	"	"	"
			125	"									"										"	$\overline{OE}$ to 3Q	"	"	"
			126	"									"										"	$\overline{OE}$ to 4Q	"	"	"
			127	"									"										"	$\overline{OE}$ to 5Q	"	"	"
			128	"									"										"	$\overline{OE}$ to 6Q	"	"	"
			129	"									"										"	$\overline{OE}$ to 7Q	"	"	"
			130	"									"										"	$\overline{OE}$ to 8Q	"	"	"
	t_{PZL}	"	131	"	OUT	GND	GND	OUT	OUT	GND	GND	OUT	"		OUT	GND	GND	OUT	OUT	GND	GND	OUT	"	$\overline{OE}$ to 1Q	6	18	"
			132	"									"										"	$\overline{OE}$ to 2Q	"	"	"
			133	"									"										"	$\overline{OE}$ to 3Q	"	"	"
			134	"									"										"	$\overline{OE}$ to 4Q	"	"	"
			135	"									"										"	$\overline{OE}$ to 5Q	"	"	"
			136	"									"										"	$\overline{OE}$ to 6Q	"	"	"
			137	"									"										"	$\overline{OE}$ to 7Q	"	"	"
			138	"									"										"	$\overline{OE}$ to 8Q	"	"	"
	t_{PHZ}	"	139	"	OUT	3.0 V	3.0 V	OUT	OUT	3.0 V	3.0 V	OUT	"		OUT	3.0 V	3.0 V	OUT	OUT	3.0 V	3.0 V	OUT	"	$\overline{OE}$ to 1Q	2	12	"
			140	"									"										"	$\overline{OE}$ to 2Q	"	"	"
			141	"									"										"	$\overline{OE}$ to 3Q	"	"	"
			142	"									"										"	$\overline{OE}$ to 4Q	"	"	"
			143	"									"										"	$\overline{OE}$ to 5Q	"	"	"
			144	"									"										"	$\overline{OE}$ to 6Q	"	"	"
			145	"									"										"	$\overline{OE}$ to 7Q	"	"	"
			146	"									"										"	$\overline{OE}$ to 8Q	"	"	"
	t_{PLZ}	"	147	"	OUT	GND	GND	OUT	OUT	GND	GND	OUT	"		OUT	GND	GND	OUT	OUT	GND	GND	OUT	"	$\overline{OE}$ to 1Q	2	10	"
			148	"									"										"	$\overline{OE}$ to 2Q	"	"	"
			149	"									"										"	$\overline{OE}$ to 3Q	"	"	"
			150	"									"										"	$\overline{OE}$ to 4Q	"	"	"
			151	"									"										"	$\overline{OE}$ to 5Q	"	"	"
			152	"									"										"	$\overline{OE}$ to 6Q	"	"	"
			153	"									"										"	$\overline{OE}$ to 7Q	"	"	"
			154	"									"										"	$\overline{OE}$ to 8Q	"	"	"
10 T_C = +125°C	t_{PLH2}	Same tests and terminal conditions as subgroup 9, except T_C = +125°C.																							2	14	"
	t_{PHL2}																								4	19	"
	t_{PLH3}																								6	26	"
	t_{PHL3}																								7	27	"
	t_{PZH}																								5	24	"
	t_{PZL}																								6	22	"
	t_{PHZ}																								2	16	"
	t_{PLZ}																								2	12	"
11 T_C = -55°C		Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																									

NOTES:

1/ I_{IL} limits shall be as follows:

Test	Min/Max limits in µA for circuits A	B	C
I_{IL}	0/-100		

2/ Method 3011 of MIL-STD-883 shall be used, except the output shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one-half of the true short-circuit output current I_{OS}.

3/ Tests shall be performed in sequence, attributes data only.
4/ A = 3.0 V minimum, B = 0.0 V or GND.
5/ H > 1.5 V., L < 1.5 V.

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TABLE III. Group A inspection for device type 04.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{OC}$	1Q	1D	2D	2Q	3Q	3D	4D	4Q	GND	CLK	5Q	5D	6D	6Q	7Q	7D	8D	8Q	V_{CC}				
1 T_C = +25°C	V_{OH}	3006	1	0.8 V	-1 mA	2.0 V	2.0 V	-1 mA	-1 mA	2.0 V	2.0 V	-1 mA	GND	1/									4.5 V	1Q	2.4		V
			2	"									"	"									"	2Q	"		"
			3	"									"	"									"	3Q	"		"
			4	"									"	"									"	4Q	"		"
			5	"									"	"									"	5Q	"		"
			6	"									"	"									"	6Q	"		"
			7	"									"	"									"	7Q	"		"
			8	"									"	"									"	8Q	"		"
	V_{OL}	3007	9	"	12 mA	0.8 V	0.8 V	12 mA	12 mA	0.8 V	0.8 V	12 mA	"	"									"	1Q		0.4	"
			10	"									"	"									"	2Q		"	"
			11	"									"	"									"	3Q		"	"
			12	"									"	"									"	4Q		"	"
			13	"									"	"									"	5Q		"	"
			14	"									"	"									"	6Q		"	"
			15	"									"	"									"	7Q		"	"
			16	"									"	"									"	8Q		"	"
	V_{IC}		17	-18 mA		-18 mA	-18 mA			-18 mA	-18 mA		"	-18 mA		-18 mA	-18 mA			-18 mA	-18 mA		"	$\overline{OC}$		-1.5	"
			18										"										"	1D		"	"
			19										"										"	2D		"	"
			20										"										"	3D		"	"
			21										"										"	4D		"	"
			22										"										"	CLK		"	"
			23										"										"	5D		"	"
			24										"										"	6D		"	"
			25										"										"	7D		"	"
			26										"										"	8D		"	"
	I_{IL}	3009	27	0.4 V		0.4 V	0.4 V			0.4 V	0.4 V		"	0.4 V		0.4 V	0.4 V			0.4 V	0.4 V		5.5 V	$\overline{OC}$	2/	2/	µA
			28										"										"	1D	"	"	"
			29										"										"	2D	"	"	"
			30										"										"	3D	"	"	"
			31										"										"	4D	"	"	"
			32										"										"	CLK	"	"	"
			33										"										"	5D	"	"	"
			34										"										"	6D	"	"	"
			35										"										"	7D	"	"	"
			36										"										"	8D	"	"	"
	I_{IH1}	3010	37	2.7 V		2.7 V	2.7 V			2.7 V	2.7 V		"	2.7 V		2.7 V	2.7 V			2.7 V	2.7 V		"	$\overline{OC}$		20	"
			38										"										"	1D		"	"
			39										"										"	2D		"	"
			40										"										"	3D		"	"
			41										"										"	4D		"	"
			42										"										"	CLK		"	"
			43										"										"	5D		"	"
			44										"										"	6D		"	"
			45										"										"	7D		"	"
			46										"										"	8D		"	"
	I_{IH2}	"	47	7.0 V		7.0 V	7.0 V			7.0 V	7.0 V		"	7.0 V		7.0 V	7.0 V			7.0 V	7.0 V		"	$\overline{OC}$		100	"
			48										"										"	1D		"	"
			49										"										"	2D		"	"
			50										"										"	3D		"	"
			51										"										"	4D		"	"
			52										"										"	CLK		"	"
			53										"										"	5D		"	"
			54										"										"	6D		"	"
			55										"										"	7D		"	"
			56										"										"	8D		"	"
	I_{OZL}		57	2.0 V	0.4 V	2.0 V	2.0 V	0.4 V	0.4 V	2.0 V	2.0 V	0.4 V	"	1/									"	1Q		-20	"
			58	"									"		0.4 V	2.0 V	2.0 V	0.4 V	0.4 V	2.0 V	2.0 V	0.4 V	"	2Q		"	"
			59	"									"										"	3Q		"	"
			60	"									"										"	4Q		"	"
			61	"									"										"	5Q		"	"
			62	"									"										"	6Q		"	"
			63	"									"										"	7Q		"	"
			64	"									"										"	8Q		"	"

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{OC}$	1Q	1D	2D	2Q	3Q	3D	4D	4Q	GND	CLK	5Q	5D	6D	6Q	7Q	7D	8D	8Q	V_{CC}				
1 T_C = +25°C	I_{OZH}		65	2.0 V	2.7 V	0.8 V							GND	1/									5.5 V	1Q		20	µA
			66	"			0.8 V	2.7 V		0.8 V			"	"									"	2Q		"	"
			67	"					2.7 V		0.8 V	2.7 V	"	"	2.7 V	0.8 V							"	3Q		"	"
			68	"									"	"			0.8 V	2.7 V					"	4Q		"	"
			69	"									"	"					2.7 V	0.8 V			"	5Q		"	"
			70	"									"	"							0.8 V	2.7 V	"	6Q		"	"
			71	"									"	"									"	7Q		"	"
			72	"									"	"									"	8Q		"	"
	I_{O} 3/	3011	73	GND	2.25 V	4.5 V							"	"									5.5 V	1Q	-30	-112	mA
		"	74				4.5 V	2.25 V		4.5 V			"	"									"	2Q	"	"	"
		"	75						2.25 V		4.5 V	2.25 V	"	"	2.25 V	4.5 V							"	3Q	"	"	"
		"	76										"	"			4.5 V	2.25 V					"	4Q	"	"	"
		"	77	GND									"	"					2.25 V	4.5 V			"	5Q	"	"	"
		"	78	"									"	"							4.5 V	2.25 V	"	6Q	"	"	"
		"	79	"									"	"									"	7Q	"	"	"
		"	80	"									"	"									"	8Q	"	"	"
	I_{CCH}	3005	81	"		4.5 V	4.5 V			4.5 V	4.5 V		"	"		4.5 V	4.5 V			4.5 V	4.5 V			"	V_{CC}		19	"
	I_{CCL}	"	82	"		GND	GND			GND	GND		"	"		GND	GND			GND	GND			"	V_{CC}		28	"
	I_{CCZ}	"	83	4.5 V		GND	GND			GND	GND		"	"		GND	GND			GND	GND			"	V_{CC}		31	"
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and omit V_{IC} tests.																										
3	Same tests, terminal conditions, and tests as subgroup 1, except T_C = -55°C and omit V_{IC} tests.																										
7 T_C = +25°C	Truth table tests 4/ 5/	3014	84	B	X 6/	A	A	X 6/	X 6/	A	A	X 6/	GND	B	X 6/	A	A	X 6/	X 6/	A	A	X 6/	4.5 V	7/			
		"	85	"	H	A	A	H	H	A	A	H	"	A	H	A	A	H	H	A	A	H	"				
		"	86	"	"	B	B	"	"	B	B	"	"	A	H	B	B	H	H	B	B	H	"				
		"	87	"	"	"	"	"	"	"	"	"	"	B	H	B	B	H	H	B	B	H	"				
		"	88	"	L	"	"	L	L	"	"	L	"	A	L	A	A	L	L	A	A	L	"				
		"	89	"	L	A	A	L	L	A	A	L	"	A	L	A	A	L	L	A	A	L	"				
		"	90	"	H	A	A	H	H	A	A	H	"	A	L	A	A	H	H	A	A	H	"				
8	Same tests and terminal conditions as subgroup 7, except T_C = +125°C and -55°C.																										
9 T_C = +25°C	f_{MAX} 8/	3003 (fig. 7)	91	GND	OUT	IN							GND	IN									5.0 V	1Q	35		MHz
		"	92	"			IN	OUT		IN			"	"									"	2Q	"		"
		"	93	"					OUT		IN	OUT	"	"	OUT	IN							"	3Q	"		"
		"	94	"									"	"			IN	OUT					"	4Q	"		"
		"	95	"									"	"					OUT	IN			"	5Q	"		"
		"	96	"									"	"							IN	OUT	"	6Q	"		"
		"	97	"									"	"									"	7Q	"		"
		"	98	"									"	"									"	8Q	"		"
	t_{PLH1}	"	99	"	OUT	IN							"	"									"	CLK > 1Q	3	12	ns
		"	100	"			IN	OUT		IN			"	"									"	CLK > 2Q	"	"	"
		"	101	"					OUT		IN	OUT	"	"	OUT	IN							"	CLK > 3Q	"	"	"
		"	102	"									"	"			IN	OUT					"	CLK > 4Q	"	"	"
		"	103	"									"	"					OUT	IN			"	CLK > 5Q	"	"	"
		"	104	"									"	"							IN	OUT	"	CLK > 6Q	"	"	"
		"	105	"									"	"									"	CLK > 7Q	"	"	"
		"	106	"									"	"									"	CLK > 8Q	"	"	"
	t_{PHL1}	"	107	"	OUT	IN							"	"									"	CLK > 1Q	5	16	"
		"	108	"			IN	OUT		IN			"	"									"	CLK > 2Q	"	"	"
		"	109	"					OUT		IN	OUT	"	"	OUT	IN							"	CLK > 3Q	"	"	"
		"	110	"									"	"			IN	OUT					"	CLK > 4Q	"	"	"
		"	111	"									"	"					OUT	IN			"	CLK > 5Q	"	"	"
		"	112	"									"	"							IN	OUT	"	CLK > 6Q	"	"	"
		"	113	"									"	"									"	CLK > 7Q	"	"	"
		"	114	"									"	"									"	CLK > 8Q	"	"	"
	t_{PZH}	"	115	IN	OUT	3.0 V							"	1/									"	$\overline{OC}$ > 1Q	5	17	"
		"	116	"			3.0 V	OUT		3.0 V			"	"									"	$\overline{OC}$ > 2Q	"	"	"
		"	117	"					OUT		3.0 V	OUT	"	"	OUT	3.0 V							"	$\overline{OC}$ > 3Q	"	"	"
		"	118	"									"	"			3.0 V	OUT					"	$\overline{OC}$ > 4Q	"	"	"
		"	119	"									"	"					OUT	3.0 V			"	$\overline{OC}$ > 5Q	"	"	"
		"	120	"									"	"							3.0 V	OUT	"	$\overline{OC}$ > 6Q	"	"	"
		"	121	"									"	"									"	$\overline{OC}$ > 7Q	"	"	"
		"	122	"									"	"									"	$\overline{OC}$ > 8Q	"	"	"

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				$\overline{OC}$	1Q	1D	2D	2Q	3Q	3D	4D	4Q	GND	CLK	5Q	5D	6D	6Q	7Q	7D	8D	8Q	V_{CC}				
9 T_C = +25°C	t_{PZL}	3003 (fig. 7)	123	IN	OUT	GND	GND	OUT	OUT	GND	GND	OUT	GND	1/	OUT	GND	GND	OUT	OUT	GND	GND	OUT	5.0 V	$\overline{OC}$ > 1Q	7	18	ns
		"	124	"									"	"									"	$\overline{OC}$ > 2Q	"	"	"
		"	125	"									"	"									"	$\overline{OC}$ > 3Q	"	"	"
		"	126	"									"	"									"	$\overline{OC}$ > 4Q	"	"	"
		"	127	"									"	"									"	$\overline{OC}$ > 5Q	"	"	"
		"	128	"									"	"									"	$\overline{OC}$ > 6Q	"	"	"
		"	129	"									"	"									"	$\overline{OC}$ > 7Q	"	"	"
		"	130	"									"	"									"	$\overline{OC}$ > 8Q	"	"	"
	t_{PHZ}	"	131	"	OUT	3.0 V	3.0 V	OUT	OUT	3.0 V	3.0 V	OUT	"	"	OUT	3.0 V	3.0 V	OUT	OUT	3.0 V	3.0 V	OUT	"	$\overline{OC}$ > 1Q	2	10	"
		"	132	"									"	"									"	$\overline{OC}$ > 2Q	"	"	"
		"	133	"									"	"									"	$\overline{OC}$ > 3Q	"	"	"
		"	134	"									"	"									"	$\overline{OC}$ > 4Q	"	"	"
		"	135	"									"	"									"	$\overline{OC}$ > 5Q	"	"	"
		"	136	"									"	"									"	$\overline{OC}$ > 6Q	"	"	"
		"	137	"									"	"									"	$\overline{OC}$ > 7Q	"	"	"
		"	138	"									"	"									"	$\overline{OC}$ > 8Q	"	"	"
	t_{PLZ}	"	139	"	OUT	GND	GND	OUT	OUT	GND	GND	OUT	"	"	OUT	GND	GND	OUT	OUT	GND	GND	OUT	"	$\overline{OC}$ > 1Q	3	18	"
		"	140	"									"	"									"	$\overline{OC}$ > 2Q	"	"	"
		"	141	"									"	"									"	$\overline{OC}$ > 3Q	"	"	"
		"	142	"									"	"									"	$\overline{OC}$ > 4Q	"	"	"
		"	143	"									"	"									"	$\overline{OC}$ > 5Q	"	"	"
		"	144	"									"	"									"	$\overline{OC}$ > 6Q	"	"	"
		"	145	"									"	"									"	$\overline{OC}$ > 7Q	"	"	"
		"	146	"									"	"									"	$\overline{OC}$ > 8Q	"	"	"
10 T_C = +125°C	f_{MAX}	Same tests and terminal conditions as subgroup 9, except T_C = +125°C.																								30	MHz
	t_{PLH1}																								3	15	ns
	t_{PHL1}																								5	18	"
	t_{PZH}																								5	19	"
	t_{PZL}																								7	20	"
	t_{PHZ}																								2	12	"
	t_{PLZ}																								3	24	"
11 T_C = -55°C		Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																									

NOTES:

1/ Apply ⌐¯¯¯¯¯¯¯¯¬ 2.0 V min/4.5 V max
└_____┘ --0 V pulse, (prior to test).

2/ I_{IL} limits shall be as follows:

Test	Min/Max limits in µA for circuits A	B	C
I_{IL}	0/-200		

3/ Method 3011 of MIL-STD-883 shall be used, except the output shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one-half of the true short-circuit output current I_{OS}.

4/ Tests shall be performed in sequence, attributes data only.

5/ A = 3.0 V, minimum, B = 0.0 V or GND.

6/ X = indeterminate output voltage.

7/ H > 1.5 V, L < 1.5 V.

8/ f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

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4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the reference terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logisitic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND - - - - - - - - - - - - - - - Ground zero voltage potential.
I_{IN} - - - - - - - - - - - - - - - Current flowing into an input terminal.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

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6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54ALS174
02	54ALS175
03	54ALS373
04	54ALS374

6.6 Manufacturers' designators. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designations.

Device type	Circuit		
	A	B	C
	Texas Instruments	Motorola Inc.	National Semiconductor Corp.
01	X		
02	X		
03	X		
04	X		

Custodian:
Air Force - 17

Review activities:
Air Force - 11, 19, 85, 99
DLA - ES

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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